

20/02/25

## MODULE II

### MATHEMATICAL LOGIC AND PROOFS

A declarative sentence which is either true or false but not both is called a proposition / statement, and it is denoted by  $p, q, r, s, t$ , etc.

If a proposition is true, then the truth value is denoted by  $T$  or  $1$ , and if it is false, it is denoted by  $F$  or  $0$ .

Example: New Delhi is the capital of India.

↳ this is a proposition.

↳ truth value is  $1$ .

$$2+3=10$$

↳ truth value is  $0$ .

Propositions which do not contain any of the logical operators or connectives is known as atomic or primary or primitive propositions.

Propositions obtained by connecting atomic propositions by logical connectives is called compound propositions.

#### LOGICAL CONNECTIVES / OPERATORS

- |                 |                   |            |
|-----------------|-------------------|------------|
| 1. CONJUNCTION  | $\wedge$          | and        |
| 2. DISJUNCTION  | $\vee$            | or         |
| 3. NEGATION     | $\neg \sim$       | not        |
| 4. EXCLUSIVE OR | $\oplus$          |            |
| 5. IMPLICATION  | $\longrightarrow$ | if... then |

## 6. BICONDITIONAL



IFF

### CONJUNCTION

If  $p$  and  $q$  are any two propositions, then the conjunction of  $p$  and  $q$  is denoted by  $p \wedge q$  and it is read as 'p and q', defined as the compound proposition that is true when both  $p$  and  $q$  are true.

| P | q | $p \wedge q$ |
|---|---|--------------|
| 1 | 1 | 1            |
| 0 | 1 | 0            |
| 1 | 0 | 0            |
| 0 | 0 | 0            |

### DISTINCTION

| P | q | $p \vee q$ |
|---|---|------------|
| 1 | 1 | 1          |
| 0 | 1 | 1          |
| 1 | 0 | 1          |
| 0 | 0 | 0          |

### NEGATION

| P | $\sim P$ |
|---|----------|
| 1 | 0        |
| 0 | 1        |

### EXCLUSIVE OR

| P | q | $p \oplus q$ |
|---|---|--------------|
| 1 | 1 | 0            |
| 0 | 1 | 1            |
| 1 | 0 | 1            |
| 0 | 0 | 0            |

### IMPLICATION $\rightarrow$ conditional

| P | q | $p \rightarrow q$ |
|---|---|-------------------|
| 1 | 1 | 1                 |
| 0 | 1 | 1                 |
| 1 | 0 | 0                 |
| 0 | 0 | 1                 |

# BICONDITIONAL

| P | q | $P \leftrightarrow q$ |
|---|---|-----------------------|
| 1 | 1 | 1                     |
| 1 | 0 | 0                     |
| 0 | 1 | 0                     |
| 0 | 0 | 1                     |

Q. Construct a truth table  $(P \vee \neg q) \rightarrow (P \wedge q)$

| P | q | $\neg q$ | $P \vee \neg q$ | $P \wedge q$ | $a \rightarrow b$ |
|---|---|----------|-----------------|--------------|-------------------|
| 1 | 1 | 0        | 1               | 1            | 1                 |
| 1 | 0 | 1        | 1               | 0            | 0                 |
| 0 | 1 | 0        | 0               | 0            | 1                 |
| 0 | 0 | 1        | 1               | 0            | 0                 |

contingency

21/02/25

Q. Construct a truth table for the compound statement  $\neg(P \vee \neg q) \rightarrow \neg P$

| P | q | $\neg q$ | $P \vee \neg q$ | $\neg(P \vee \neg q)$ | $\neg P$ | $a \rightarrow b$ |
|---|---|----------|-----------------|-----------------------|----------|-------------------|
| 1 | 1 | 0        | 1               | 0                     | 0        | 1                 |
| 0 | 1 | 0        | 0               | 1                     | 1        | 1                 |
| 1 | 0 | 1        | 1               | 0                     | 0        | 1                 |
| 0 | 0 | 1        | 1               | 0                     | 1        | 1                 |

tautology

Q. Find the truth table of  $p \wedge (\neg p \wedge q)$

| P | q | $\neg p$ | $\neg p \wedge q$ | $p \wedge (\neg p \wedge q)$ |
|---|---|----------|-------------------|------------------------------|
| 1 | 1 | 0        | 0                 | 0                            |
| 0 | 1 | 1        | 1                 | 0                            |
| 1 | 0 | 0        | 0                 | 0                            |
| 0 | 0 | 1        | 0                 | 0                            |

fallacy  
contradiction.

**TAUTOLOGY:** A compound proposition is called tautology if it is true for all possible assignments for its primitive statements.

**CONTRADICTION:** If a compound proposition is false for all such assignments, then it is called a contradiction.

**CONTINGENCY:** A compound proposition that is neither tautology nor contradiction is called contingency.

Q. Construct a truth table for  $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ .  
check whether it is a tautology.

| P | q | $p \rightarrow q$ | $\neg q$ | $\neg p$ | $\neg q \rightarrow \neg p$ | $a \leftrightarrow b$ |
|---|---|-------------------|----------|----------|-----------------------------|-----------------------|
| 1 | 1 | 1                 | 0        | 0        | 1                           | 1                     |
| 0 | 1 | 1                 | 0        | 1        | 1                           | 1                     |
| 1 | 0 | 0                 | 1        | 0        | 0                           | 1                     |
| 0 | 0 | 1                 | 1        | 1        | 1                           | 1                     |

It is a tautology



Q. construct a truth table for  $(p \vee q) \rightarrow (p \oplus q)$

|   |   | $\overset{a}{p \vee q}$ |              | $\overset{b}{p \oplus q}$ | $a \rightarrow b$ |
|---|---|-------------------------|--------------|---------------------------|-------------------|
| P | q | $p \vee q$              | $p \oplus q$ |                           |                   |
| 1 | 1 | 1                       | 0            | 0                         | 0                 |
| 0 | 1 | 1                       | 1            | 1                         | 1                 |
| 1 | 0 | 1                       | 1            | 1                         | 1                 |
| 0 | 0 | 0                       | 0            | 0                         | 1                 |

Q. check whether the implication  $p \rightarrow (q \rightarrow r)$  is a tautology by constructing the truth table.

| P | q | r | $q \rightarrow r$ | $p \rightarrow (q \rightarrow r)$ |
|---|---|---|-------------------|-----------------------------------|
| 0 | 0 | 0 | 1                 | 1                                 |
| 0 | 0 | 1 | 1                 | 1                                 |
| 0 | 1 | 0 | 0                 | 1                                 |
| 1 | 0 | 0 | 1                 | 0                                 |
| 1 | 1 | 0 | 0                 | 0                                 |
| 0 | 1 | 1 | 1                 | 1                                 |
| 1 | 0 | 1 | 1                 | 0                                 |
| 1 | 1 | 1 | 1                 | 1                                 |

It is not a tautology

Q.  $q \leftrightarrow (\neg p \vee \neg q)$

| P | q | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ | $q \leftrightarrow (\neg p \vee \neg q)$ |
|---|---|----------|----------|----------------------|------------------------------------------|
| 1 | 1 | 0        | 0        | 0                    | 0                                        |
| 1 | 0 | 0        | 1        | 1                    | 0                                        |
| 0 | 1 | 1        | 0        | 1                    | 1                                        |
| 0 | 0 | 1        | 1        | 1                    | 0                                        |

Q1.  $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg x) \vee q) \leftarrow (x \wedge p) \wedge (p \leftrightarrow q)$

| $p$ | $q$ | $x$ | $p \leftrightarrow q$ | $\neg p$ | $\neg x$ | $\neg p \leftrightarrow \neg x$ | $a \oplus b$ |
|-----|-----|-----|-----------------------|----------|----------|---------------------------------|--------------|
| 1   | 1   | 1   | 1                     | 0        | 0        | 1                               | 0            |
| 1   | 1   | 0   | 1                     | 0        | 1        | 0                               | 1            |
| 1   | 0   | 1   | 0                     | 0        | 0        | 1                               | 1            |
| 1   | 0   | 0   | 0                     | 0        | 1        | 0                               | 0            |
| 0   | 1   | 1   | 0                     | 1        | 0        | 0                               | 0            |
| 0   | 1   | 0   | 0                     | 1        | 1        | 1                               | 1            |
| 0   | 0   | 1   | 1                     | 1        | 0        | 0                               | 1            |
| 0   | 0   | 0   | 1                     | 1        | 1        | 0                               | 0            |

Q2.  $(\neg(p \wedge q) \vee x) \rightarrow \neg p$

| $p$ | $q$ | $x$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg(p \wedge q) \vee x$ | $\neg p$ | $a \rightarrow b$ |
|-----|-----|-----|--------------|--------------------|---------------------------|----------|-------------------|
| 1   | 1   | 1   | 1            | 0                  | 1                         | 0        | 0                 |
| 1   | 1   | 0   | 1            | 0                  | 0                         | 0        | 1                 |
| 1   | 0   | 1   | 0            | 1                  | 1                         | 0        | 0                 |
| 1   | 0   | 0   | 0            | 1                  | 1                         | 0        | 0                 |
| 0   | 1   | 1   | 0            | 1                  | 1                         | 1        | 1                 |
| 0   | 1   | 0   | 0            | 1                  | 1                         | 1        | 1                 |
| 0   | 0   | 1   | 0            | 1                  | 1                         | 1        | 1                 |
| 0   | 0   | 0   | 0            | 1                  | 1                         | 1        | 1                 |

Q3.  $(p \rightarrow q) \wedge [(q \wedge \neg x) \rightarrow (p \vee x)] \leftrightarrow (p \leftrightarrow q)$

|   |   |   | a                 |          | b                 |            | c                 |                              |  |
|---|---|---|-------------------|----------|-------------------|------------|-------------------|------------------------------|--|
| p | q | x | $p \rightarrow q$ | $\neg x$ | $q \wedge \neg x$ | $p \vee x$ | $b \rightarrow c$ | $a \wedge (b \rightarrow c)$ |  |
| 1 | 1 | 1 | 1                 | 0        | 0                 | 1          | 1                 | 1                            |  |
| 1 | 1 | 0 | 1                 | 1        | 1                 | 1          | 1                 | 1                            |  |
| 1 | 0 | 1 | 0                 | 0        | 0                 | 1          | 1                 | 0                            |  |
| 1 | 0 | 0 | 0                 | 1        | 0                 | 1          | 1                 | 0                            |  |
| 0 | 1 | 1 | 1                 | 0        | 0                 | 1          | 1                 | 1                            |  |
| 0 | 1 | 0 | 1                 | 1        | 1                 | 0          | 0                 | 0                            |  |
| 0 | 0 | 1 | 1                 | 0        | 0                 | 1          | 1                 | 1                            |  |
| 0 | 0 | 0 | 1                 | 1        | 0                 | 0          | 1                 | 1                            |  |

$\neg \left( (x \vee (p \wedge q)) \right)$

Q4.  $(p \rightarrow q) \rightarrow (x \rightarrow s)$

| p | q | x | s | $p \rightarrow q$ | $x \rightarrow s$ | $(p \rightarrow q) \rightarrow (x \rightarrow s)$ |
|---|---|---|---|-------------------|-------------------|---------------------------------------------------|
| 1 | 1 | 1 | 1 | 1                 | 1                 | 1                                                 |
| 1 | 1 | 1 | 0 | 1                 | 0                 | 0                                                 |
| 1 | 1 | 0 | 1 | 1                 | 1                 | 1                                                 |
| 1 | 1 | 0 | 0 | 1                 | 1                 | 1                                                 |
| 1 | 0 | 1 | 1 | 0                 | 1                 | 0                                                 |
| 1 | 0 | 1 | 0 | 0                 | 0                 | 1                                                 |
| 1 | 0 | 0 | 1 | 0                 | 1                 | 1                                                 |
| 1 | 0 | 0 | 0 | 0                 | 1                 | 1                                                 |
| 0 | 1 | 1 | 1 | 1                 | 1                 | 1                                                 |
| 0 | 1 | 1 | 0 | 1                 | 0                 | 0                                                 |
| 0 | 1 | 0 | 1 | 1                 | 1                 | 1                                                 |
| 0 | 1 | 0 | 0 | 1                 | 1                 | 1                                                 |
| 0 | 0 | 1 | 1 | 1                 | 1                 | 1                                                 |
| 0 | 0 | 1 | 0 | 1                 | 0                 | 0                                                 |
| 0 | 0 | 0 | 1 | 1                 | 1                 | 1                                                 |
| 0 | 0 | 0 | 0 | 1                 | 1                 | 1                                                 |

# LOGICAL EQUIVALENCE

Two statements  $S_1$  and  $S_2$  are said to be logically equivalent, we write  $S_1 \equiv S_2$ . The statements  $S_1$  and  $S_2$  provide the same truth tables for all choices for their primitive components.  $(S_1 \Leftrightarrow S_2)$

Q. ST  $(P \rightarrow q)$  and  $(\neg P \vee q)$  are logically equivalent.

$P \rightarrow q$

| P | q | $P \rightarrow q$ |
|---|---|-------------------|
| 1 | 1 | 1                 |
| 1 | 0 | 0                 |
| 0 | 1 | 1                 |
| 0 | 0 | 1                 |

$\neg P \vee q$

| P | q | $\neg P$ | $\neg P \vee q$ |
|---|---|----------|-----------------|
| 1 | 1 | 0        | 1               |
| 1 | 0 | 0        | 0               |
| 0 | 1 | 1        | 1               |
| 0 | 0 | 1        | 1               |

$\therefore P \rightarrow q$  and  $\neg P \vee q$  are logically equivalent.

Q. check whether the propositions  $P \wedge (\neg q \vee r)$  and  $P \vee (q \wedge \neg r)$  are logically equivalent

| P | q | r | $\neg q$ | $\neg r$ | $\neg q \vee r$ | $q \wedge \neg r$ | $P \wedge b$ | $P \vee c$ |
|---|---|---|----------|----------|-----------------|-------------------|--------------|------------|
| 1 | 1 | 1 | 0        | 0        | 1               | 0                 | 1            | 1          |
| 1 | 1 | 0 | 0        | 1        | 0               | 1                 | 0            | 1          |
| 1 | 0 | 1 | 1        | 0        | 1               | 0                 | 1            | 1          |
| 1 | 0 | 0 | 1        | 1        | 1               | 0                 | 1            | 1          |
| 0 | 1 | 1 | 0        | 0        | 1               | 0                 | 0            | 0          |
| 0 | 1 | 0 | 0        | 1        | 0               | 1                 | 0            | 1          |
| 0 | 0 | 1 | 1        | 0        | 1               | 0                 | 0            | 0          |
| 0 | 0 | 0 | 1        | 1        | 1               | 0                 | 0            | 0          |

Hence, they are not logically equivalent.



Q1. check  $p \vee (q \wedge r)$  and  $(p \vee q) \wedge (p \vee r)$  are  $\equiv$

| P | q | r | $q \wedge r$ | $p \vee q$ | $p \vee r$ | $p \vee (q \wedge r)$ | $(p \vee q) \wedge (p \vee r)$ |
|---|---|---|--------------|------------|------------|-----------------------|--------------------------------|
| 1 | 1 | 1 | 1            | 1          | 1          | 1                     | 1                              |
| 1 | 1 | 0 | 0            | 1          | 1          | 1                     | 1                              |
| 1 | 0 | 1 | 0            | 1          | 1          | 1                     | 1                              |
| 1 | 0 | 0 | 0            | 1          | 0          | 1                     | 0                              |
| 0 | 1 | 1 | 1            | 1          | 1          | 1                     | 1                              |
| 0 | 1 | 0 | 0            | 1          | 0          | 1                     | 0                              |
| 0 | 0 | 1 | 0            | 0          | 1          | 0                     | 0                              |
| 0 | 0 | 0 | 0            | 0          | 0          | 0                     | 0                              |

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

Q2. check  $p \rightarrow (q \rightarrow r)$  and  $(p \rightarrow q) \rightarrow r$  are  $\equiv$

| P | q | r | $q \rightarrow r$ | $p \rightarrow (q \rightarrow r)$ | $p \rightarrow q$ | $(p \rightarrow q) \rightarrow r$ |
|---|---|---|-------------------|-----------------------------------|-------------------|-----------------------------------|
| 1 | 1 | 1 | 1                 | 1                                 | 1                 | 1                                 |
| 1 | 1 | 0 | 0                 | 0                                 | 1                 | 0                                 |
| 1 | 0 | 1 | 1                 | 1                                 | 0                 | 1                                 |
| 1 | 0 | 0 | 1                 | 1                                 | 0                 | 1                                 |
| 0 | 1 | 1 | 1                 | 1                                 | 1                 | 1                                 |
| 0 | 1 | 0 | 0                 | 1                                 | 1                 | 0                                 |
| 0 | 0 | 1 | 1                 | 1                                 | 1                 | 1                                 |
| 0 | 0 | 0 | 1                 | 1                                 | 1                 | 1                                 |

$p \rightarrow (q \rightarrow r)$  and  $(p \rightarrow q) \rightarrow r$  are not logically equivalent.

Q3. check  $[P \rightarrow (q \vee r)] \Leftrightarrow [\neg r \rightarrow (P \rightarrow q)]$

|   |   |   | a          |                   | b        |                   |                        |
|---|---|---|------------|-------------------|----------|-------------------|------------------------|
| P | q | r | $q \vee r$ | $P \rightarrow a$ | $\neg r$ | $P \rightarrow q$ | $\neg r \rightarrow b$ |
| 1 | 1 | 1 | 1          | 1                 | 0        | 1                 | 1                      |
| 1 | 1 | 0 | 1          | 1                 | 1        | 1                 | 1                      |
| 1 | 0 | 1 | 1          | 1                 | 0        | 0                 | 1                      |
| 1 | 0 | 0 | 0          | 0                 | 1        | 0                 | 0                      |
| 0 | 1 | 1 | 1          | 1                 | 0        | 1                 | 1                      |
| 0 | 1 | 0 | 1          | 1                 | 1        | 1                 | 1                      |
| 0 | 0 | 1 | 1          | 1                 | 0        | 1                 | 1                      |
| 0 | 0 | 0 | 0          | 1                 | 1        | 1                 | 1                      |

$$[P \rightarrow (q \vee r)] \Leftrightarrow [\neg r \rightarrow (P \rightarrow q)]$$

Hence, proved

Q4. check  $P \vee (q \wedge r)$  and  $(P \vee q) \wedge r$

|   |   |   | a            |            | b          |              |
|---|---|---|--------------|------------|------------|--------------|
| P | q | r | $q \wedge r$ | $P \vee a$ | $P \vee q$ | $b \wedge r$ |
| 1 | 1 | 1 | 1            | 1          | 1          | 1            |
| 1 | 1 | 0 | 0            | 1          | 1          | 0            |
| 1 | 0 | 1 | 0            | 1          | 1          | 0            |
| 1 | 0 | 0 | 0            | 1          | 1          | 0            |
| 0 | 1 | 1 | 1            | 1          | 1          | 1            |
| 0 | 1 | 0 | 0            | 0          | 1          | 0            |
| 0 | 0 | 1 | 0            | 0          | 0          | 0            |
| 0 | 0 | 0 | 0            | 0          | 0          | 0            |

$P \vee (q \wedge r)$  and  $(P \vee q) \wedge r$  are not VE.

33. Construct a truth table for each of these compound propositions.

a)  $p \wedge \neg p$

| p | $\neg p$ | $p \wedge \neg p$ |
|---|----------|-------------------|
| 1 | 0        | 0                 |
| 0 | 1        | 0                 |

(b)  $p \vee \neg p$

| p | $\neg p$ | $p \vee \neg p$ |
|---|----------|-----------------|
| 1 | 0        | 1               |
| 0 | 1        | 1               |

(c)  $(p \vee \neg q) \rightarrow q$

| p | q | $\neg q$ | $p \vee \neg q$ | $(p \vee \neg q) \rightarrow q$ |
|---|---|----------|-----------------|---------------------------------|
| 1 | 1 | 0        | 1               | 1                               |
| 1 | 0 | 1        | 1               | 0                               |
| 0 | 1 | 0        | 0               | 1                               |
| 0 | 0 | 1        | 1               | 0                               |

(d)  $(p \vee q) \rightarrow (p \wedge q)$

| p | q | $p \vee q$ | $p \wedge q$ | $(p \vee q) \rightarrow (p \wedge q)$ |
|---|---|------------|--------------|---------------------------------------|
| 1 | 1 | 1          | 1            | 1                                     |
| 1 | 0 | 1          | 0            | 0                                     |
| 0 | 1 | 1          | 0            | 0                                     |
| 0 | 0 | 0          | 0            | 1                                     |

(e)  $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

| p | q | $p \rightarrow q$ | $\neg p$ | $\neg q$ | $\neg q \rightarrow \neg p$ | $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ |
|---|---|-------------------|----------|----------|-----------------------------|-----------------------------------------------------------------|
| 1 | 1 | 1                 | 0        | 0        | 1                           | 1                                                               |
| 1 | 0 | 0                 | 0        | 1        | 0                           | 1                                                               |
| 0 | 1 | 1                 | 1        | 0        | 1                           | 1                                                               |
| 0 | 0 | 1                 | 1        | 1        | 1                           | 1                                                               |

$$(f) (p \rightarrow q) \rightarrow (q \rightarrow p)$$

| p | q | $p \rightarrow q$ | $q \rightarrow p$ | $p \rightarrow q \rightarrow p$ |
|---|---|-------------------|-------------------|---------------------------------|
| 1 | 1 | 1                 | 1                 | 1                               |
| 1 | 0 | 0                 | 1                 | 0                               |
| 0 | 1 | 1                 | 0                 | 0                               |
| 0 | 0 | 1                 | 1                 | 1                               |

$$34. a) p \rightarrow \neg p$$

| p | $\neg p$ | $p \rightarrow \neg p$ |
|---|----------|------------------------|
| 1 | 0        | 0                      |
| 0 | 1        | 1                      |

$$b) p \leftrightarrow \neg p$$

| p | $\neg p$ | $p \leftrightarrow \neg p$ |
|---|----------|----------------------------|
| 1 | 0        | 0                          |
| 0 | 1        | 0                          |

$$(c) p \oplus (p \vee q)$$

| p | q | $p \vee q$ | $p \oplus (p \vee q)$ |
|---|---|------------|-----------------------|
| 1 | 1 | 1          | 0                     |
| 1 | 0 | 1          | 0                     |
| 0 | 1 | 1          | 1                     |
| 0 | 0 | 0          | 0                     |

$$d) p \wedge q \rightarrow p \vee q$$

| p | q | $p \wedge q$ | $p \vee q$ | $p \wedge q \rightarrow p \vee q$ |
|---|---|--------------|------------|-----------------------------------|
| 1 | 1 | 1            | 1          | 1                                 |
| 1 | 0 | 0            | 1          | 1                                 |
| 0 | 1 | 0            | 1          | 1                                 |
| 0 | 0 | 0            | 0          | 1                                 |

$$e) (q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$$

| p | q | $\neg p$ | $q \rightarrow \neg p$ | $p \leftrightarrow q$ | $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$ |
|---|---|----------|------------------------|-----------------------|----------------------------------------------------------------|
| 1 | 1 | 0        | 0                      | 1                     | 0                                                              |
| 1 | 0 | 0        | 1                      | 0                     | 0                                                              |
| 0 | 1 | 1        | 1                      | 0                     | 1                                                              |
| 0 | 0 | 1        | 1                      | 1                     | 0                                                              |

$$f) (p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$$

| p | q | $p \leftrightarrow q$ | $\neg q$ | $p \leftrightarrow \neg q$ | $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$ |
|---|---|-----------------------|----------|----------------------------|-----------------------------------------------------------|
| 1 | 1 | 1                     | 0        | 0                          | 1                                                         |
| 1 | 0 | 0                     | 1        | 1                          | 0                                                         |
| 0 | 1 | 0                     | 0        | 0                          | 0                                                         |
| 0 | 0 | 1                     | 1        | 1                          | 0                                                         |



35. a)  $(p \vee q) \rightarrow (p \oplus q)$

| p | q | $p \vee q$ | $p \oplus q$ | $a \rightarrow b$ |
|---|---|------------|--------------|-------------------|
| 1 | 1 | 1          | 0            | 0                 |
| 1 | 0 | 1          | 1            | 1                 |
| 0 | 1 | 1          | 1            | 1                 |
| 0 | 0 | 0          | 0            | 1                 |

| $(q \rightarrow p) \leftrightarrow (p \rightarrow q)$ | p | q |
|-------------------------------------------------------|---|---|
| 1                                                     | 1 | 1 |
| 0                                                     | 0 | 1 |
| 1                                                     | 1 | 0 |
| 0                                                     | 0 | 0 |

b)  $(p \oplus q) \rightarrow (p \wedge q)$

| p | q | $p \oplus q$ | $p \wedge q$ | $a \rightarrow b$ |
|---|---|--------------|--------------|-------------------|
| 1 | 1 | 0            | 1            | 1                 |
| 1 | 0 | 1            | 0            | 0                 |
| 0 | 1 | 1            | 0            | 0                 |
| 0 | 0 | 0            | 0            | 1                 |

| $q \rightarrow p$ | q | p |
|-------------------|---|---|
| 0                 | 0 | 1 |
| 1                 | 1 | 0 |

(c)  $(p \vee q) \oplus (p \wedge q)$

| p | q | $p \vee q$ | $p \wedge q$ | $a \oplus b$ |
|---|---|------------|--------------|--------------|
| 1 | 1 | 1          | 1            | 0            |
| 1 | 0 | 1          | 0            | 1            |
| 0 | 1 | 1          | 0            | 1            |
| 0 | 0 | 0          | 0            | 0            |

| $p \oplus q$ | $p \vee q$ | p | q |
|--------------|------------|---|---|
| 0            | 1          | 1 | 1 |
| 0            | 1          | 0 | 1 |
| 1            | 1          | 1 | 0 |
| 0            | 0          | 0 | 0 |

d)  $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg q)$

| p | q | $\neg p$ | $\neg q$ | $p \leftrightarrow q$ | $\neg p \leftrightarrow \neg q$ | $a \oplus b$ |
|---|---|----------|----------|-----------------------|---------------------------------|--------------|
| 1 | 1 | 0        | 0        | 1                     | 1                               | 0            |
| 1 | 0 | 0        | 1        | 0                     | 0                               | 0            |
| 0 | 1 | 1        | 0        | 0                     | 0                               | 0            |
| 0 | 0 | 1        | 1        | 1                     | 1                               | 0            |

$(p \leftrightarrow q) \leftrightarrow (\neg p \leftrightarrow \neg q)$

| $p \leftrightarrow q$ | $\neg p \leftrightarrow \neg q$ | p | q |
|-----------------------|---------------------------------|---|---|
| 1                     | 1                               | 1 | 1 |
| 0                     | 0                               | 1 | 0 |
| 0                     | 0                               | 0 | 1 |
| 1                     | 1                               | 0 | 0 |

d)  $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$

| p | q | $\neg p$ | $p \leftrightarrow q$ | $\neg p \leftrightarrow q$ | $a \oplus b$ |
|---|---|----------|-----------------------|----------------------------|--------------|
| 1 | 1 | 0        | 1                     | 0                          | 1            |
| 1 | 0 | 0        | 0                     | 1                          | 1            |
| 0 | 1 | 1        | 0                     | 1                          | 1            |
| 0 | 0 | 1        | 1                     | 0                          | 1            |

f)  $(p \oplus q) \rightarrow (p \oplus \neg q)$

| p | q | $\neg q$ | $p \oplus q$ | $p \oplus \neg q$ | $a \rightarrow b$ |
|---|---|----------|--------------|-------------------|-------------------|
| 1 | 1 | 0        | 0            | 1                 | 1                 |
| 1 | 0 | 1        | 1            | 0                 | 0                 |
| 0 | 1 | 0        | 1            | 0                 | 1                 |
| 0 | 0 | 1        | 0            | 1                 | 1                 |

36 a)  $p \oplus p$

| p | $p \oplus p$ |
|---|--------------|
| 1 | 0            |
| 0 | 0            |

b)  $p \oplus \neg p$

| p | $\neg p$ | $p \oplus \neg p$ |
|---|----------|-------------------|
| 1 | 0        | 1                 |
| 0 | 1        | 1                 |

c)  $p \oplus \neg q$

| p | q | $\neg q$ | $p \oplus \neg q$ |
|---|---|----------|-------------------|
| 1 | 1 | 0        | 1                 |
| 1 | 0 | 1        | 0                 |
| 0 | 1 | 0        | 0                 |
| 0 | 0 | 1        | 1                 |

(d)  $\neg p \oplus \neg q$

| p | q | $\neg p$ | $\neg q$ | $\neg p \oplus \neg q$ |
|---|---|----------|----------|------------------------|
| 1 | 1 | 0        | 0        | 0                      |
| 1 | 0 | 0        | 1        | 1                      |
| 0 | 1 | 1        | 0        | 1                      |
| 0 | 0 | 1        | 1        | 0                      |

e)  $(p \oplus q) \vee (p \oplus \neg q)$

| p | q | $\neg q$ | $p \oplus q$ | $p \oplus \neg q$ | $a \vee b$ |
|---|---|----------|--------------|-------------------|------------|
| 1 | 1 | 0        | 0            | 1                 | 1          |
| 1 | 0 | 1        | 1            | 0                 | 1          |
| 0 | 1 | 0        | 1            | 0                 | 1          |
| 0 | 0 | 1        | 0            | 1                 | 1          |

(f)  $(p \oplus q) \wedge (p \oplus \neg q)$

| p | q | $\neg q$ | $p \oplus q$ | $p \oplus \neg q$ | $a \wedge b$ |
|---|---|----------|--------------|-------------------|--------------|
| 1 | 1 | 0        | 0            | 1                 | 0            |
| 1 | 0 | 1        | 1            | 0                 | 0            |
| 0 | 1 | 0        | 1            | 0                 | 0            |
| 0 | 0 | 1        | 0            | 1                 | 0            |

$$37-a) P \rightarrow \neg q$$

| P | q | $\neg q$ | $P \rightarrow \neg q$ |
|---|---|----------|------------------------|
| 1 | 1 | 0        | 0                      |
| 1 | 0 | 1        | 1                      |
| 0 | 1 | 0        | 1                      |
| 0 | 0 | 1        | 1                      |

$$b) \neg P \leftrightarrow q$$

| P | q | $\neg P$ | $\neg P \leftrightarrow q$ |
|---|---|----------|----------------------------|
| 1 | 1 | 0        | 0                          |
| 1 | 0 | 0        | 0                          |
| 0 | 1 | 1        | 1                          |
| 0 | 0 | 1        | 0                          |

$$c) (P \rightarrow q) \vee (\neg P \rightarrow q)$$

| P | q | $\neg P$ | $P \xrightarrow{a} q$ | $\neg P \xrightarrow{b} q$ | $a \vee b$ | $a \wedge b$ |
|---|---|----------|-----------------------|----------------------------|------------|--------------|
| 1 | 1 | 0        | 1                     | 1                          | 1          | 1            |
| 1 | 0 | 0        | 0                     | 1                          | 1          | 0            |
| 0 | 1 | 1        | 1                     | 0                          | 1          | 0            |
| 0 | 0 | 1        | 1                     | 0                          | 1          | 0            |

$$d) (P \rightarrow q) \wedge (\neg P \rightarrow q) = (P \oplus q)$$

$$e) (P \leftrightarrow q) \vee (\neg P \leftrightarrow q)$$

| P | q | $P \xleftrightarrow{a} q$ | $\neg P \xleftrightarrow{b} q$ | $a \vee b$ | $a \oplus b$ |
|---|---|---------------------------|--------------------------------|------------|--------------|
| 1 | 1 | 1                         | 0                              | 1          | 0            |
| 1 | 0 | 0                         | 1                              | 1          | 1            |
| 0 | 1 | 0                         | 0                              | 0          | 1            |
| 0 | 0 | 1                         | 1                              | 1          | 0            |

$$(f) (\neg P \leftrightarrow \neg q) \leftrightarrow (P \leftrightarrow q)$$

| P | q | $\neg P$ | $\neg q$ | $\neg P \xleftrightarrow{a} \neg q$ | $P \xleftrightarrow{b} q$ | $a \leftrightarrow b$ |
|---|---|----------|----------|-------------------------------------|---------------------------|-----------------------|
| 1 | 1 | 0        | 0        | 1                                   | 1                         | 1                     |
| 1 | 0 | 0        | 1        | 0                                   | 0                         | 1                     |
| 0 | 1 | 1        | 0        | 1                                   | 1                         | 1                     |
| 0 | 0 | 1        | 1        | 0                                   | 0                         | 1                     |

28. a)  $(p \vee q) \vee r$       b)  $(p \vee q) \wedge r$

| $p$ | $q$ | $r$ | $(p \vee q)$ | $a \vee r$ | $a \wedge r$ | $(p \wedge q)$ | $(p \wedge q) \wedge r$ |
|-----|-----|-----|--------------|------------|--------------|----------------|-------------------------|
| 1   | 1   | 1   | 1            | 1          | 1            | 1              | 1                       |
| 1   | 1   | 0   | 1            | 1          | 0            | 1              | 0                       |
| 1   | 0   | 1   | 1            | 1          | 0            | 0              | 0                       |
| 1   | 0   | 0   | 1            | 1          | 0            | 0              | 0                       |
| 0   | 1   | 1   | 1            | 1          | 0            | 0              | 0                       |
| 0   | 1   | 0   | 1            | 1          | 0            | 0              | 0                       |
| 0   | 0   | 1   | 0            | 0          | 0            | 0              | 0                       |
| 0   | 0   | 0   | 0            | 0          | 0            | 0              | 0                       |

c)  $(p \wedge q) \vee r$       d)  $(p \wedge q) \wedge r$

| $p$ | $q$ | $r$ | $p \wedge q$ | $a \vee r$ | $a \wedge r$ | $p \vee r$ | $(r \vee p) \wedge q$ | $(r \vee p) \wedge r$ |
|-----|-----|-----|--------------|------------|--------------|------------|-----------------------|-----------------------|
| 1   | 1   | 1   | 1            | 1          | 1            | 1          | 1                     | 1                     |
| 1   | 1   | 0   | 1            | 0          | 0            | 0          | 0                     | 0                     |
| 1   | 0   | 1   | 0            | 1          | 0            | 1          | 0                     | 0                     |
| 1   | 0   | 0   | 0            | 0          | 0            | 1          | 0                     | 0                     |
| 0   | 1   | 1   | 0            | 1          | 0            | 1          | 0                     | 0                     |
| 0   | 1   | 0   | 0            | 0          | 0            | 0          | 0                     | 0                     |
| 0   | 0   | 1   | 0            | 1          | 0            | 1          | 0                     | 0                     |
| 0   | 0   | 0   | 0            | 0          | 0            | 1          | 0                     | 0                     |

e)  $(p \vee q) \wedge r$

| $p$ | $q$ | $r$ | $p \vee q$ | $r$ | $a \wedge r$ | $(r \leftarrow p) \leftarrow q$ | $(r \leftarrow p) \leftarrow q \wedge r$ |
|-----|-----|-----|------------|-----|--------------|---------------------------------|------------------------------------------|
| 1   | 1   | 1   | 1          | 1   | 1            | 1                               | 1                                        |
| 1   | 1   | 0   | 1          | 0   | 0            | 0                               | 0                                        |
| 1   | 0   | 1   | 1          | 1   | 1            | 0                               | 0                                        |
| 1   | 0   | 0   | 1          | 0   | 0            | 0                               | 0                                        |
| 0   | 1   | 1   | 1          | 1   | 1            | 1                               | 1                                        |
| 0   | 1   | 0   | 1          | 0   | 0            | 0                               | 0                                        |
| 0   | 0   | 1   | 0          | 1   | 0            | 1                               | 0                                        |
| 0   | 0   | 0   | 0          | 0   | 0            | 0                               | 0                                        |



$$f) (p \wedge q) \vee \neg r$$

| p | q | r |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 0 | 0 |
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |
| 0 | 0 | 0 |

| $\neg p$ | $\neg q$ | $\neg r$ | $(p \wedge q)$ | $(p \wedge q) \vee \neg r$ |
|----------|----------|----------|----------------|----------------------------|
| 0        | 0        | 1        | 1              | 1                          |
| 0        | 1        | 1        | 1              | 1                          |
| 0        | 1        | 0        | 0              | 0                          |
| 0        | 0        | 1        | 0              | 1                          |
| 1        | 0        | 1        | 0              | 1                          |
| 1        | 0        | 0        | 0              | 0                          |
| 1        | 1        | 1        | 0              | 1                          |
| 1        | 1        | 0        | 0              | 0                          |

| $\neg r$ | $p$ | $q$ | $(p \wedge q)$ | $(p \wedge q) \vee \neg r$ |
|----------|-----|-----|----------------|----------------------------|
| 1        | 1   | 1   | 1              | 1                          |
| 1        | 1   | 0   | 0              | 1                          |
| 1        | 0   | 1   | 0              | 1                          |
| 1        | 0   | 0   | 0              | 1                          |
| 0        | 1   | 1   | 1              | 1                          |
| 0        | 1   | 0   | 0              | 0                          |
| 0        | 0   | 1   | 0              | 0                          |
| 0        | 0   | 0   | 0              | 0                          |

| $\neg r$ | $p$ | $q$ | $(p \wedge q)$ | $(p \wedge q) \vee \neg r$ |
|----------|-----|-----|----------------|----------------------------|
| 1        | 1   | 1   | 1              | 1                          |
| 1        | 1   | 0   | 0              | 1                          |
| 1        | 0   | 1   | 0              | 1                          |
| 1        | 0   | 0   | 0              | 1                          |
| 0        | 1   | 1   | 1              | 1                          |
| 0        | 1   | 0   | 0              | 0                          |
| 0        | 0   | 1   | 0              | 0                          |
| 0        | 0   | 0   | 0              | 0                          |

$$g) a) p \rightarrow (\neg q \vee r)$$

| p | q | r | $\neg q$ |
|---|---|---|----------|
| 1 | 1 | 1 | 0        |
| 1 | 1 | 0 | 0        |
| 1 | 0 | 1 | 1        |
| 1 | 0 | 0 | 1        |
| 0 | 1 | 1 | 0        |
| 0 | 1 | 0 | 0        |
| 0 | 0 | 1 | 1        |
| 0 | 0 | 0 | 1        |

| $\neg q \vee r$ | $p \rightarrow (\neg q \vee r)$ |
|-----------------|---------------------------------|
| 1               | 1                               |
| 0               | 0                               |
| 1               | 1                               |
| 1               | 1                               |
| 1               | 1                               |
| 0               | 0                               |
| 1               | 1                               |
| 1               | 1                               |

| $p \rightarrow (\neg q \vee r)$ |
|---------------------------------|
| 1                               |
| 0                               |
| 1                               |
| 1                               |
| 1                               |
| 0                               |
| 1                               |
| 1                               |

| $p \rightarrow (\neg q \vee r)$ |
|---------------------------------|
| 1                               |
| 0                               |
| 1                               |
| 1                               |
| 1                               |
| 0                               |
| 1                               |
| 1                               |

$$b) \neg p \rightarrow (q \rightarrow r)$$

| p | q | r | $\neg p$ |
|---|---|---|----------|
| 1 | 1 | 1 | 0        |
| 1 | 1 | 0 | 0        |
| 1 | 0 | 1 | 0        |
| 1 | 0 | 0 | 0        |
| 0 | 1 | 1 | 1        |
| 0 | 1 | 0 | 1        |
| 0 | 0 | 1 | 1        |
| 0 | 0 | 0 | 1        |

| $q \rightarrow r$ |
|-------------------|
| 1                 |
| 0                 |
| 1                 |
| 0                 |
| 1                 |
| 0                 |
| 1                 |
| 1                 |

| $\neg p \rightarrow (q \rightarrow r)$ |
|----------------------------------------|
| 1                                      |
| 1                                      |
| 1                                      |
| 1                                      |
| 1                                      |
| 0                                      |
| 0                                      |
| 1                                      |

| $\neg p \rightarrow (q \rightarrow r)$ |
|----------------------------------------|
| 1                                      |
| 1                                      |
| 1                                      |
| 1                                      |
| 1                                      |
| 0                                      |
| 0                                      |
| 1                                      |

c)  $(p \rightarrow q) \vee (\neg p \rightarrow r)$       d)  $(p \rightarrow q) \wedge (\neg p \rightarrow r)$

| p | q | r | $\neg p$ | $p \xrightarrow{a} q$ | $\neg p \xrightarrow{b} r$ | $a \vee b$ | $a \wedge b$ |
|---|---|---|----------|-----------------------|----------------------------|------------|--------------|
| 1 | 1 | 1 | 0        | 1                     | 1                          | 1          | 1            |
| 1 | 1 | 0 | 0        | 1                     | 1                          | 1          | 1            |
| 1 | 0 | 1 | 0        | 0                     | 1                          | 1          | 0            |
| 1 | 0 | 0 | 0        | 0                     | 1                          | 1          | 0            |
| 0 | 1 | 1 | 1        | 1                     | 1                          | 1          | 1            |
| 0 | 1 | 0 | 1        | 1                     | 1                          | 1          | 1            |
| 0 | 0 | 1 | 1        | 1                     | 0                          | 1          | 0            |
| 0 | 0 | 0 | 1        | 1                     | 0                          | 1          | 0            |

e)  $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$

| p | q | r | $\neg q$ | $p \xleftrightarrow{a} q$ | $\neg q \xleftrightarrow{b} r$ | $a \vee b$ |
|---|---|---|----------|---------------------------|--------------------------------|------------|
| 1 | 1 | 1 | 0        | 1                         | 0                              | 1          |
| 1 | 1 | 0 | 0        | 1                         | 1                              | 1          |
| 1 | 0 | 1 | 1        | 0                         | 1                              | 1          |
| 1 | 0 | 0 | 1        | 0                         | 0                              | 0          |
| 0 | 1 | 1 | 0        | 0                         | 0                              | 0          |
| 0 | 1 | 0 | 0        | 0                         | 1                              | 1          |
| 0 | 0 | 1 | 1        | 1                         | 1                              | 1          |
| 0 | 0 | 0 | 1        | 1                         | 0                              | 1          |

f)  $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

| p | q | r | $\neg p$ | $\neg q$ | $\neg p \xleftrightarrow{a} \neg q$ | $q \xleftrightarrow{b} r$ | $a \leftrightarrow b$ |
|---|---|---|----------|----------|-------------------------------------|---------------------------|-----------------------|
| 1 | 1 | 1 | 0        | 0        | 1                                   | 1                         | 1                     |
| 1 | 1 | 0 | 0        | 0        | 1                                   | 0                         | 0                     |
| 1 | 0 | 1 | 0        | 1        | 0                                   | 0                         | 1                     |
| 1 | 0 | 0 | 0        | 1        | 0                                   | 1                         | 0                     |
| 0 | 1 | 1 | 1        | 0        | 1                                   | 1                         | 1                     |
| 0 | 1 | 0 | 1        | 0        | 1                                   | 0                         | 0                     |
| 0 | 0 | 1 | 1        | 1        | 1                                   | 0                         | 0                     |
| 0 | 0 | 0 | 1        | 1        | 1                                   | 1                         | 1                     |

40. Construct a truth table for  $((p \rightarrow q) \rightarrow r) \rightarrow s$

$$p \rightarrow q \quad r \rightarrow s \quad p \rightarrow q \quad a \rightarrow b \quad b \rightarrow c \quad r \rightarrow p \quad q \rightarrow r$$

| $p$ | $q$ | $r$ | $s$ | $p \rightarrow q$ | $a \rightarrow b$ | $b \rightarrow c$ | $r \rightarrow p$ | $q \rightarrow r$ |
|-----|-----|-----|-----|-------------------|-------------------|-------------------|-------------------|-------------------|
| 1   | 1   | 1   | 1   | 1                 | 1                 | 1                 | 1                 | 1                 |
| 1   | 1   | 1   | 0   | 1                 | 1                 | 0                 | 0                 | 1                 |
| 1   | 1   | 0   | 1   | 1                 | 0                 | 1                 | 0                 | 0                 |
| 1   | 1   | 0   | 0   | 1                 | 0                 | 1                 | 0                 | 0                 |
| 1   | 0   | 1   | 1   | 0                 | 1                 | 1                 | 0                 | 0                 |
| 1   | 0   | 1   | 0   | 0                 | 0                 | 1                 | 0                 | 0                 |
| 1   | 0   | 0   | 1   | 0                 | 1                 | 1                 | 0                 | 0                 |
| 1   | 0   | 0   | 0   | 0                 | 1                 | 0                 | 0                 | 0                 |
| 0   | 1   | 1   | 1   | 1                 | 1                 | 1                 | 1                 | 1                 |
| 0   | 1   | 1   | 0   | 1                 | 1                 | 0                 | 0                 | 1                 |
| 0   | 1   | 0   | 1   | 1                 | 0                 | 1                 | 0                 | 0                 |
| 0   | 1   | 0   | 0   | 1                 | 0                 | 1                 | 0                 | 0                 |
| 0   | 0   | 1   | 1   | 1                 | 1                 | 1                 | 1                 | 1                 |
| 0   | 0   | 1   | 0   | 1                 | 1                 | 0                 | 0                 | 1                 |
| 0   | 0   | 0   | 1   | 1                 | 0                 | 1                 | 0                 | 0                 |
| 0   | 0   | 0   | 0   | 1                 | 0                 | 0                 | 0                 | 0                 |

41.  $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$

$(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$   
 $p \quad q \quad r \quad s \quad p \leftrightarrow q \quad r \leftrightarrow s \quad (p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 |

23. show that  $\neg p \leftrightarrow q$  and  $p \leftrightarrow \neg q$  are logically equi.

| p | q | $\neg p$ | $\neg q$ | $\neg p \leftrightarrow q$ | $p \leftrightarrow \neg q$ |
|---|---|----------|----------|----------------------------|----------------------------|
| 1 | 1 | 0        | 0        | 0                          | 0                          |
| 1 | 0 | 0        | 1        | 1                          | 1                          |
| 0 | 1 | 1        | 0        | 1                          | 1                          |
| 0 | 0 | 1        | 1        | 0                          | 0                          |

$\neg p \leftrightarrow q$  and  $p \leftrightarrow \neg q$  are logically equivalent.

Hence, proved

24. ST  $\neg(p \oplus q)$  and  $p \leftrightarrow q$  are LE.

| p | q | $p \oplus q$ | $\neg(p \oplus q)$ | $p \leftrightarrow q$ |
|---|---|--------------|--------------------|-----------------------|
| 1 | 1 | 0            | 1                  | 1                     |
| 1 | 0 | 1            | 0                  | 0                     |
| 0 | 1 | 1            | 0                  | 0                     |
| 0 | 0 | 0            | 1                  | 1                     |

$\neg(p \oplus q)$  and  $p \leftrightarrow q$  are logically equivalent.

Hence, proved

25. ST  $\neg(p \leftrightarrow q)$  and  $\neg p \leftrightarrow q$  are LE.

| p | q | $p \leftrightarrow q$ | $\neg(p \leftrightarrow q)$ | $\neg p$ | $\neg p \leftrightarrow q$ |
|---|---|-----------------------|-----------------------------|----------|----------------------------|
| 1 | 1 | 1                     | 0                           | 0        | 0                          |
| 1 | 0 | 0                     | 1                           | 0        | 0                          |
| 0 | 1 | 0                     | 1                           | 1        | 1                          |
| 0 | 0 | 1                     | 0                           | 1        | 0                          |

$\neg(p \leftrightarrow q)$  and  $\neg p \leftrightarrow q$  are LE.

26. Hence, proved

proving, unit



26. ST  $(p \rightarrow q) \wedge (p \rightarrow r)$  and  $p \rightarrow (q \wedge r)$  are LE.

| $p$ | $q$ | $r$ | $p \xrightarrow{a} q$ | $p \xrightarrow{b} r$ | $a \wedge b$ | $q \xrightarrow{c} r$ | $p \rightarrow c$ |
|-----|-----|-----|-----------------------|-----------------------|--------------|-----------------------|-------------------|
| 1   | 1   | 1   | 1                     | 1                     | 1            | 1                     | 1                 |
| 1   | 1   | 0   | 1                     | 0                     | 0            | 0                     | 0                 |
| 1   | 0   | 1   | 0                     | 1                     | 0            | 0                     | 0                 |
| 1   | 0   | 0   | 0                     | 0                     | 0            | 0                     | 0                 |
| 0   | 1   | 1   | 1                     | 1                     | 1            | 1                     | 1                 |
| 0   | 1   | 0   | 1                     | 0                     | 0            | 0                     | 0                 |
| 0   | 0   | 1   | 1                     | 1                     | 1            | 0                     | 0                 |
| 0   | 0   | 0   | 1                     | 1                     | 1            | 0                     | 0                 |

$(p \rightarrow q) \wedge (p \rightarrow r)$  and  $p \rightarrow (q \wedge r)$  are ~~not~~ LE.

Hence, proved

27. ST  $(p \rightarrow r) \wedge (q \rightarrow r)$  and  $(p \vee q) \rightarrow r$  are LE.

| $p$ | $q$ | $r$ | $p \xrightarrow{a} r$ | $q \xrightarrow{b} r$ | $a \wedge b$ | $p \vee q$ | $c \xrightarrow{c} r$ |
|-----|-----|-----|-----------------------|-----------------------|--------------|------------|-----------------------|
| 1   | 1   | 1   | 1                     | 1                     | 1            | 1          | 1                     |
| 1   | 1   | 0   | 0                     | 0                     | 0            | 1          | 0                     |
| 1   | 0   | 1   | 1                     | 0                     | 0            | 1          | 1                     |
| 1   | 0   | 0   | 0                     | 0                     | 0            | 1          | 0                     |
| 0   | 1   | 1   | 1                     | 1                     | 1            | 1          | 1                     |
| 0   | 1   | 0   | 0                     | 0                     | 0            | 1          | 0                     |
| 0   | 0   | 1   | 1                     | 1                     | 1            | 0          | 1                     |
| 0   | 0   | 0   | 1                     | 1                     | 1            | 0          | 0                     |

$(p \rightarrow r) \wedge (q \rightarrow r)$  and  $(p \vee q) \rightarrow r$  are LE.

Hence, proved

28. ST  $(p \rightarrow q) \vee (p \rightarrow r)$  and  $p \rightarrow (q \vee r)$  are t.e. T2

| p | q | r | $p \xrightarrow{a} q$ | $p \xrightarrow{b} r$ | $a \vee b$ | $q \vee r$ | $p \rightarrow c$ |
|---|---|---|-----------------------|-----------------------|------------|------------|-------------------|
| 1 | 1 | 1 | 1                     | 1                     | 1          | 1          | 1                 |
| 1 | 1 | 0 | 1                     | 0                     | 1          | 1          | 1                 |
| 1 | 0 | 1 | 0                     | 1                     | 1          | 1          | 1                 |
| 1 | 0 | 0 | 0                     | 0                     | 0          | 0          | 0                 |
| 0 | 1 | 1 | 1                     | 1                     | 1          | 1          | 1                 |
| 0 | 1 | 0 | 1                     | 0                     | 1          | 1          | 1                 |
| 0 | 0 | 1 | 0                     | 1                     | 1          | 1          | 1                 |
| 0 | 0 | 0 | 0                     | 0                     | 0          | 0          | 0                 |

$(p \rightarrow q) \vee (p \rightarrow r)$  and  $p \rightarrow (q \vee r)$  are t.e.

Hence, proved

29. ST  $(p \rightarrow r) \vee (q \rightarrow r)$  and  $(p \wedge q) \rightarrow r$  are t.e.

| p | q | r | $p \xrightarrow{a} r$ | $q \xrightarrow{b} r$ | $a \vee b$ | $p \wedge q \xrightarrow{c} r$ |
|---|---|---|-----------------------|-----------------------|------------|--------------------------------|
| 1 | 1 | 1 | 1                     | 1                     | 1          | 1                              |
| 1 | 1 | 0 | 0                     | 0                     | 0          | 0                              |
| 1 | 0 | 1 | 1                     | 0                     | 1          | 1                              |
| 1 | 0 | 0 | 0                     | 0                     | 0          | 0                              |
| 0 | 1 | 1 | 1                     | 1                     | 1          | 1                              |
| 0 | 1 | 0 | 0                     | 1                     | 1          | 1                              |
| 0 | 0 | 1 | 1                     | 0                     | 1          | 1                              |
| 0 | 0 | 0 | 0                     | 0                     | 0          | 0                              |

$(p \rightarrow r) \vee (q \rightarrow r)$  and  $(p \wedge q) \rightarrow r$  are t.e.

Hence, proved

| p | q | $p \rightarrow q$ | p | q |
|---|---|-------------------|---|---|
| 1 | 1 | 1                 | 1 | 1 |
| 1 | 0 | 0                 | 1 | 0 |
| 0 | 1 | 1                 | 0 | 1 |
| 0 | 0 | 1                 | 0 | 0 |

30. ST  $\neg p \rightarrow (q \rightarrow r)$  and  $q \rightarrow (p \vee r)$  are tautologies (T.E.)

| p | q | r | $\neg p$ | $q \rightarrow r$ | $\neg p \rightarrow (q \rightarrow r)$ | $p \vee r$ | $q \rightarrow (p \vee r)$ |
|---|---|---|----------|-------------------|----------------------------------------|------------|----------------------------|
| 1 | 1 | 1 | 0        | 1                 | 1                                      | 1          | 1                          |
| 1 | 1 | 0 | 0        | 0                 | 0                                      | 1          | 0                          |
| 1 | 0 | 1 | 0        | 1                 | 1                                      | 1          | 1                          |
| 1 | 0 | 0 | 0        | 1                 | 1                                      | 1          | 1                          |
| 0 | 1 | 1 | 1        | 1                 | 1                                      | 1          | 1                          |
| 0 | 1 | 0 | 1        | 0                 | 0                                      | 1          | 1                          |
| 0 | 0 | 1 | 1        | 1                 | 1                                      | 1          | 1                          |
| 0 | 0 | 0 | 1        | 1                 | 1                                      | 0          | 1                          |

$\neg p \rightarrow (q \rightarrow r)$  and  $q \rightarrow (p \vee r)$  are tautologies (T.E.)

Hence, proved

31. ST  $p \leftrightarrow q$  and  $(p \rightarrow q) \wedge (q \rightarrow p)$  are tautologies (T.E.)

| p | q | $p \leftrightarrow q$ | $p \rightarrow q$ | $q \rightarrow p$ | $(p \rightarrow q) \wedge (q \rightarrow p)$ |
|---|---|-----------------------|-------------------|-------------------|----------------------------------------------|
| 1 | 1 | 1                     | 1                 | 1                 | 1                                            |
| 1 | 0 | 0                     | 0                 | 1                 | 0                                            |
| 0 | 1 | 0                     | 1                 | 0                 | 0                                            |
| 0 | 0 | 1                     | 1                 | 1                 | 1                                            |

$p \leftrightarrow q$  and  $(p \rightarrow q) \wedge (q \rightarrow p)$  are tautologies (T.E.)

Hence, proved

32. ST  $p \leftrightarrow q$  and  $\neg p \leftrightarrow \neg q$  are tautologies (T.E.)

| p | q | $p \leftrightarrow q$ | $\neg p$ | $\neg q$ | $\neg p \leftrightarrow \neg q$ |
|---|---|-----------------------|----------|----------|---------------------------------|
| 1 | 1 | 1                     | 0        | 0        | 1                               |
| 1 | 0 | 0                     | 0        | 1        | 0                               |
| 0 | 1 | 0                     | 1        | 0        | 0                               |
| 0 | 0 | 1                     | 1        | 1        | 1                               |

$p \leftrightarrow q$  and  $\neg p \leftrightarrow \neg q$  are tautologies.

Hence, proved

33. ST  $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$  is a tautology.

|   |   |   | <sup>a</sup><br>$p \rightarrow q$ | <sup>b</sup><br>$q \rightarrow r$ | <sup>c</sup><br>$a \wedge b$ | <sup>d</sup><br>$p \rightarrow r$ | $c \rightarrow d$ |
|---|---|---|-----------------------------------|-----------------------------------|------------------------------|-----------------------------------|-------------------|
| P | q | r |                                   |                                   |                              |                                   |                   |
| 1 | 1 | 1 | 1                                 | 1                                 | 1                            | 1                                 | 1                 |
| 1 | 1 | 0 | 1                                 | 0                                 | 0                            | 0                                 | 1                 |
| 1 | 0 | 1 | 0                                 | 1                                 | 0                            | 1                                 | 1                 |
| 1 | 0 | 0 | 0                                 | 1                                 | 0                            | 0                                 | 1                 |
| 0 | 1 | 1 | 1                                 | 1                                 | 1                            | 1                                 | 1                 |
| 0 | 1 | 0 | 1                                 | 0                                 | 0                            | 1                                 | 1                 |
| 0 | 0 | 1 | 1                                 | 1                                 | 1                            | 1                                 | 1                 |
| 0 | 0 | 0 | 1                                 | 1                                 | 1                            | 1                                 | 1                 |

Hence, proved

Therefore,  $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$  is a tautology

Hence, proved

34. ST  $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$  is a tautology.

|   |   |   | <sup>a</sup><br>$p \vee q$ | <sup>b</sup><br>$\neg p \vee r$ | <sup>d</sup><br>$a \wedge b$ | <sup>c</sup><br>$q \vee r$ | $d \rightarrow c$ |
|---|---|---|----------------------------|---------------------------------|------------------------------|----------------------------|-------------------|
| P | q | r |                            |                                 |                              |                            |                   |
| 1 | 1 | 1 | 1                          | 0                               | 0                            | 1                          | 1                 |
| 1 | 1 | 0 | 1                          | 0                               | 0                            | 1                          | 1                 |
| 1 | 0 | 1 | 1                          | 0                               | 0                            | 1                          | 1                 |
| 1 | 0 | 0 | 1                          | 0                               | 0                            | 0                          | 1                 |
| 0 | 1 | 1 | 1                          | 1                               | 1                            | 1                          | 1                 |
| 0 | 1 | 0 | 1                          | 1                               | 1                            | 1                          | 1                 |
| 0 | 0 | 1 | 0                          | 1                               | 0                            | 1                          | 1                 |
| 0 | 0 | 0 | 0                          | 1                               | 0                            | 0                          | 1                 |

$\therefore (p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$  is a tautology.

Hence, proved





37. ST  $(p \rightarrow q) \rightarrow (r \rightarrow s)$  and  $(p \rightarrow r) \rightarrow (q \rightarrow s)$  are not UE

| p | q | r | s | $p \xrightarrow{a} q$ | $r \xrightarrow{b} s$ | $a \rightarrow b$ | $p \xrightarrow{c} r$ | $q \xrightarrow{d} s$ | $c \rightarrow d$ |
|---|---|---|---|-----------------------|-----------------------|-------------------|-----------------------|-----------------------|-------------------|
| 1 | 1 | 1 | 1 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 1 | 1 | 1 | 0 | 1                     | 0                     | 0                 | 1                     | 0                     | 0                 |
| 1 | 1 | 0 | 1 | 1                     | 1                     | 1                 | 0                     | 1                     | 1                 |
| 1 | 1 | 0 | 0 | 1                     | 1                     | 1                 | 0                     | 0                     | 1                 |
| 1 | 0 | 1 | 1 | 0                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 1 | 0 | 1 | 0 | 0                     | 0                     | 1                 | 1                     | 1                     | 1                 |
| 1 | 0 | 0 | 1 | 0                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 1 | 0 | 0 | 0 | 0                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 0 | 1 | 1 | 1 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 0 | 1 | 1 | 0 | 1                     | 0                     | 0                 | 1                     | 0                     | 0                 |
| 0 | 1 | 0 | 1 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 0 | 1 | 0 | 0 | 1                     | 1                     | 1                 | 1                     | 0                     | 0                 |
| 0 | 0 | 1 | 1 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 0 | 0 | 1 | 0 | 1                     | 0                     | 0                 | 1                     | 1                     | 1                 |
| 0 | 0 | 0 | 1 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |
| 0 | 0 | 0 | 0 | 1                     | 1                     | 1                 | 1                     | 1                     | 1                 |

$\therefore (p \rightarrow q) \rightarrow (r \rightarrow s)$  and  $(p \rightarrow r) \rightarrow (q \rightarrow s)$  are not UE.

Hence, proved

$$\begin{aligned}
 & \equiv (r \vee p) \vee q \\
 & (r \vee q) \vee (p \vee q) \\
 & \equiv (r \vee p) \vee q \\
 & (r \vee q) \vee (p \vee q)
 \end{aligned}$$

$$q \equiv q \vee q$$

$$p \vee q \equiv q$$

$$p \vee q \equiv p$$

$$p \vee q \equiv p$$

03/03/25

## LAWS OF LOGIC

let  $p, q, r$  be any primitive statements and  $T_0$  is tautology and  $F_0$  is the contradiction.

|    |                                                                                                                          |                        |
|----|--------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1. | $\neg(\neg p) \equiv p$                                                                                                  | law of double negation |
| 2. | $\neg(p \vee q) \equiv \neg p \wedge \neg q$<br>$\neg(p \wedge q) \equiv \neg p \vee \neg q$                             | De-Morgan's law        |
| 3. | $p \vee q \equiv q \vee p$<br>$p \wedge q \equiv q \wedge p$                                                             | commutative laws       |
| 4. | $p \vee (q \vee r) \equiv (p \vee q) \vee r$<br>$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$                     | Associative law        |
| 5. | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$<br>$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | Distributive law       |
| 6. | $p \wedge p \equiv p$<br>$p \vee p \equiv p$                                                                             | Idempotent law.        |
| 7. | $p \vee F_0 \equiv p$<br>$p \wedge T_0 \equiv p$                                                                         | Identity laws.         |

|     |                                                               |                  |
|-----|---------------------------------------------------------------|------------------|
| 8.  | $P \vee \neg P \equiv T_0$ $P \wedge \neg P \equiv F_0$       | Negation laws    |
| 9.  | $P \wedge F_0 \equiv F_0$ $P \vee T_0 \equiv T_0$             | Domination laws  |
| 10. | $P \vee (P \wedge Q) \equiv P$ $P \wedge (P \vee Q) \equiv P$ | Absorption laws. |

### LOGICAL EQUIVALENCE INVOLVING CONDITIONAL STATEMENTS

- $P \rightarrow Q \equiv \neg P \vee Q$
- $P \rightarrow Q \equiv \neg Q \rightarrow \neg P$
- $P \wedge Q \equiv \neg(P \rightarrow \neg Q)$
- $P \vee Q \equiv \neg P \rightarrow Q$
- $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$

### LOGICAL EQUIVALENCE INVOLVING BICONDITIONAL STATEMENTS

- $P \oplus Q \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$
- $P \leftrightarrow Q \equiv \neg P \leftrightarrow \neg Q$
- $P \leftrightarrow Q \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$
- $\neg(P \leftrightarrow Q) \equiv P \leftrightarrow \neg Q$

$$(p \vee q) \wedge (p \vee \neg q) \equiv (p \wedge p) \vee q$$

$$(p \wedge q) \vee (p \wedge \neg q) \equiv (p \wedge p) \wedge q$$



Q. State and prove De-Morgan's law.

$$\textcircled{1} \neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\textcircled{2} \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\textcircled{1} \neg(p \vee q) \equiv \neg p \wedge \neg q$$

| P | q | $p \vee q$ | $\neg(p \vee q)$ | $\neg p$ | $\neg q$ | $\neg p \wedge \neg q$ |
|---|---|------------|------------------|----------|----------|------------------------|
| 1 | 1 | 1          | 0                | 0        | 0        | 0                      |
| 1 | 0 | 1          | 0                | 0        | 1        | 0                      |
| 0 | 1 | 1          | 0                | 1        | 0        | 0                      |
| 0 | 0 | 0          | 1                | 1        | 1        | 1                      |

Hence, proved

$$\textcircled{2} \neg(p \wedge q) \equiv \neg p \vee \neg q$$

| P | q | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|---|---|--------------|--------------------|----------|----------|----------------------|
| 1 | 1 | 1            | 0                  | 0        | 0        | 0                    |
| 1 | 0 | 0            | 1                  | 0        | 1        | 1                    |
| 0 | 1 | 0            | 1                  | 1        | 0        | 1                    |
| 0 | 0 | 0            | 1                  | 1        | 1        | 1                    |

Hence, proved

Q. Prove distributive law.

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

| $p$ | $q$ | $r$ | $q \wedge r$ | $p \vee q$ | $p \vee r$ | $p \vee (q \wedge r)$ | $(p \vee q) \wedge (p \vee r)$ |
|-----|-----|-----|--------------|------------|------------|-----------------------|--------------------------------|
| 1   | 1   | 1   | 1            | 1          | 1          | 1                     | 1                              |
| 1   | 1   | 0   | 0            | 1          | 1          | 1                     | 1                              |
| 1   | 0   | 1   | 0            | 1          | 1          | 1                     | 1                              |
| 1   | 0   | 0   | 0            | 1          | 1          | 1                     | 1                              |
| 0   | 1   | 1   | 1            | 1          | 1          | 1                     | 1                              |
| 0   | 1   | 0   | 0            | 1          | 1          | 1                     | 1                              |
| 0   | 0   | 1   | 0            | 0          | 1          | 0                     | 0                              |
| 0   | 0   | 0   | 0            | 0          | 0          | 0                     | 0                              |

Hence,  $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ , proved

CONVERSE, INVERSE & CONTRAPOSITIVE

If  $p \rightarrow q$  is an implication, then,

converse is  $q \rightarrow p$

inverse is  $\neg p \rightarrow \neg q$

contrapositive is  $\neg q \rightarrow \neg p$

converse

inverse

contra positive

| $p$ | $q$ | $p \rightarrow q$ | $q \rightarrow p$ | $\neg p \rightarrow \neg q$ | $\neg p \rightarrow \neg q$ | $\neg q \rightarrow \neg p$ |
|-----|-----|-------------------|-------------------|-----------------------------|-----------------------------|-----------------------------|
| 1   | 1   | 1                 | 1                 | 0                           | 0                           | 1                           |
| 1   | 0   | 0                 | 1                 | 0                           | 1                           | 0                           |
| 0   | 1   | 1                 | 0                 | 1                           | 0                           | 1                           |
| 0   | 0   | 1                 | 1                 | 1                           | 1                           | 1                           |

$$p \rightarrow q \not\equiv q \rightarrow p$$

$$p \rightarrow q \not\equiv \neg p \rightarrow \neg q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Q. Write the converse, inverse and contrapositive.

1. If today is Friday, then tomorrow is Saturday.

$p$ : Today is Friday

$q$ : Tomorrow is Saturday

Given statement is  $p \rightarrow q$

Converse:  $q \rightarrow p$

If tomorrow is Saturday, then today is Friday.

Inverse:  $\neg p \rightarrow \neg q$

If today is not Friday, then tomorrow is not Saturday.

Contrapositive:  $\neg q \rightarrow \neg p$

If tomorrow is not Saturday, then today is not Friday.

2. If  $P$  is a square, then  $P$  is a rectangle.

$p$ :  $P$  is a square

$q$ :  $P$  is a rectangle

Given statement is  $p \rightarrow q$

Converse:  $q \rightarrow p$

If  $P$  is a rectangle, then  $P$  is a square.

Inverse: If  $P$  is not a square, then  $P$  is not a rectangle.

Contrapositive: If  $P$  is not a rectangle, then  $P$  is not a square.

3. If  $0+0=0$ , then  $1+1=1$ . Also find the truth value.

$p$ :  $0+0=0$  (True)

$q$ :  $1+1=1$  (False)

Given statement is  $P \rightarrow Q$

$\Rightarrow T \rightarrow F$ , then F

converse,  $Q \rightarrow P$

$\Rightarrow F \rightarrow T$ , then T

If  $1+1=1$ , then  $0+0=0$

inverse,  $\neg P \rightarrow \neg Q$

$\Rightarrow F \rightarrow T$ , then T

If  $0+0 \neq 0$ , then  $1+1 \neq 1$ .

contrapositive,  $\neg Q \rightarrow \neg P$

$\Rightarrow T \rightarrow F$  then F

If  $1+1 \neq 1$ , then  $0+0 \neq 0$ .

HW.

04/03/25

Q. Write the converse, inverse and contrapositive of the following statements.

a) If it snows today, I will ski tomorrow.

P: It snows today

Q: I will ski tomorrow.

Given statement is  $P \rightarrow Q$

Converse :  $Q \rightarrow P$

If I ski tomorrow, then it will snow today.

Inverse : If it does not snow today, I will not ski tomorrow

$\neg P \rightarrow \neg Q$

Contrapositive :  $\neg Q \rightarrow \neg P$

If I don't ski tomorrow, then it will not snow today.

b) A positive integer is a prime only if it has no divisors other than 1 and itself.

P: A positive integer is a prime

Q: A positive integer has no divisors other than 1 and itself.

The given statement is  $P \leftrightarrow Q$ .



Converse:  $\neg q \rightarrow \neg p$  If a positive integer has no other divisors other than 1 and itself then it is a prime.

Inverse:  $\neg p \rightarrow \neg q$  If a number is not a prime, then it has other divisors other than 1 and itself.

Contrapositive:  $\neg q \rightarrow \neg p$ , If a number has divisors other than 1 and itself then it is not a prime.

05/09/25

Q. Show that  $\neg(p \rightarrow q)$  and  $(p \wedge \neg q)$  are LE.

① by using truth table method.

② w/o using truth table

①

| p | q | $p \rightarrow q$ | $\neg p$ | $\neg q$ | $p \wedge \neg q$ |
|---|---|-------------------|----------|----------|-------------------|
| 1 | 1 | 1                 | 0        | 0        | 0                 |
| 1 | 0 | 0                 | 0        | 1        | 1                 |
| 0 | 1 | 1                 | 1        | 0        | 0                 |
| 0 | 0 | 1                 | 1        | 1        | 0                 |

Hence,  $\neg(p \rightarrow q) \equiv (p \wedge \neg q)$

②  $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ ,  $p \rightarrow q \equiv \neg p \vee q$

$\equiv \neg(\neg p) \wedge \neg q$ , De-Morgan's law

$\equiv \underline{\underline{p \wedge \neg q}}$ , double negation law

Q. PT  $[(p \vee q) \wedge (p \vee \neg q)] \vee q$  is LE to  $p \vee q$ .

$$[(p \vee q) \wedge (p \vee \neg q)] \vee q \equiv [p \vee (q \wedge \neg q)] \vee q, \text{ distributive law}$$

$$\equiv [p \vee (F_0)] \vee q, \text{ negation law}$$

$$\equiv \underline{\underline{p \vee q}}, \text{ identity law}$$

LHS  $\equiv$  RHS

Hence, proved

Q. ST  $\neg(p \vee (\neg p \wedge q))$  and  $\neg p \wedge \neg q$  are LE, by developing a series of logical equivalences.

$$\text{LHS} \equiv \neg(p \vee (\neg p \wedge q)) \equiv \neg((p \vee \neg p) \wedge (p \vee q)), \text{ distributive law}$$

$$\equiv \neg(T_0 \wedge (p \vee q)), \text{ negation law}$$

$$\equiv \neg(p \vee q), \text{ identity law}$$

$$\equiv \neg p \wedge \neg q, \text{ de-morgan's law}$$

$$\equiv \text{RHS}$$

LHS  $\equiv$  RHS, Hence, proved

$$\text{Q. PT } (p \vee q) \wedge \neg(\neg p \wedge q) \equiv p$$

$$\text{LHS} \equiv (p \vee q) \wedge \neg(\neg p \wedge q), \text{ de-morgan's law}$$

$$\equiv (p \vee q) \wedge [\neg(\neg p) \vee \neg q], \text{ de-morgan's law}$$

$$\equiv (p \vee q) \wedge (p \vee \neg q), \text{ double negation law}$$

$$\equiv p \vee (q \wedge \neg q), \text{ distributive law}$$

$$\equiv p \vee F_0, \text{ negation law}$$

$$\equiv p, \text{ identity law}$$

$$\equiv \text{RHS}, \underline{\underline{\text{hence proved}}}$$

Q. PT  $\neg P \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$ .

LHS  $\equiv \neg P \rightarrow (q \rightarrow r)$

$\equiv \neg P \rightarrow (\neg q \vee r)$ ,  $P \rightarrow Q \equiv \neg P \vee Q$

$\equiv \neg(\neg P) \vee (\neg q \vee r)$ ,  $P \rightarrow Q \equiv \neg P \vee Q$

$\equiv P \vee (\neg q \vee r)$ , double negation law

$\equiv \cancel{P \vee (r \vee \neg q)}$

$\equiv \neg q \vee (P \vee r)$ , associative law

$\equiv \neg q \rightarrow (P \vee r)$ ,  $P \vee Q \equiv \neg P \rightarrow Q$

$\equiv$  RHS, hence, proved

Q. PT  $(P \rightarrow Q) \wedge (R \rightarrow Q) \equiv (P \vee R) \rightarrow Q$ .

LHS  $\equiv (P \rightarrow Q) \wedge (R \rightarrow Q) \equiv (\neg P \vee Q) \wedge (\neg R \vee Q)$ ,

$P \rightarrow Q \equiv \neg P \vee Q$

$\equiv (\neg P \vee Q) \wedge (\neg R \vee Q)$ , commutative law

$\equiv Q \vee (\neg P \wedge \neg R)$ , distributive law

$\equiv (\neg P \wedge \neg R) \vee Q$ , comm. law.

$\equiv \neg(P \vee R) \vee Q$ , de-morgan's law

$\equiv (P \vee R) \rightarrow Q$ ,  $\neg P \vee Q \equiv P \rightarrow Q$

$\equiv$  RHS, hence, proved

Q. PT  $(P \wedge Q) \rightarrow (P \vee Q)$  is a tautology.



$$\begin{aligned}
 (p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q), \quad p \rightarrow q \equiv \neg p \vee q \\
 &\equiv (\neg p \vee \neg q) \vee (p \vee q), \text{ de-morgan's law} \\
 &\equiv (\neg p \vee p) \vee (\neg q \vee q), \text{ associative law \& commu.} \\
 &\equiv T_0 \vee T_0, \text{ negation law} \\
 &\equiv \underline{T_0}, \text{ hence, proved it is a tautology}
 \end{aligned}$$

Q. write the negation and express in words:

① If it is raining, then the game is cancelled.

p: It is raining

q: The game is cancelled.

The given statement is  $p \rightarrow q$

$$\begin{aligned}
 \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q), \quad p \rightarrow q \equiv \neg p \vee q \\
 &\equiv \neg(\neg p) \wedge \neg q, \text{ double negation} \\
 &\equiv \underline{p \wedge \neg q}
 \end{aligned}$$

Therefore, the negation is:

It is raining and the game is not cancelled.

② If the milk is white, then the cow is not black.

p: The milk is white

q: The cow is ~~not~~ black.

The given statement is  $p \rightarrow \neg q$

$$\begin{aligned}
 \neg(p \rightarrow \neg q) &\equiv p \wedge \neg(\neg q) \\
 &\equiv \underline{p \wedge q}, \text{ double negation law.}
 \end{aligned}$$

Therefore the negation is,

The milk is white and the cow is black.



③ If he studies, he will pass the examination. (p → q)

p: He studies.

q: He passes the examination.

Given statement is  $p \rightarrow q$ .

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

The negation is,

He studies ~~and~~ <sup>but</sup> he will not pass the examination.

④ If I drive, then I will not walk.

p: I drive

q: I will walk.

Given statement is  $p \rightarrow \neg q$

$$\neg(p \rightarrow \neg q) \equiv p \wedge q$$

I drive and I will walk.

⑤ Paris is in France and London is in England.

p: Paris is in France

q: London is in England

Given:  $p \wedge q$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q, \text{ de-morgan's law.}$$

Ans:- Paris is not in France or London is not in England.

⑥ He swims if and only if the water is warm.

p: he swims

q: the water is warm

Given:  $p \leftrightarrow q$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

He swims if and only if the water is not warm.

⑦  $a$  is greater than 8 and 6 is less than 10.

$p: a > 8$

$q: 6 < 10$ .

Given:  $p \wedge q$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$ , De Morgan's law.

Ans:  ~~$a$  is not greater than 8 or 6 is not greater than 8 or 6 is less than or equal to 8 or 8 is greater than equal to 10.~~

$a \leq 8$  or  $6 \geq 10$ .

RULE OF INFERENCE  $\rightarrow$  premise  $\frac{1}{2}$  conclusion.

Q. Rita is baking a cake. If Rita is baking a cake, then she is not practising her flute. If Rita is not practicing her flute, then her father will not buy her a car. Therefore, Rita's father will not buy a car. Check the validity of the argument.

$p$ : Rita is baking a cake.  $\neg(p \wedge q) \equiv$

$q$ : <sup>Rita</sup> she is practicing her flute.  $\neg(p \wedge q) \equiv$

$r$ : Rita's father will buy her a car.  $\neg(p \wedge q) \equiv$

The argument is

$$\begin{array}{l} p \\ p \rightarrow \neg q \\ \neg q \rightarrow \neg r \end{array} \left\{ \begin{array}{l} \text{premise} \end{array} \right.$$

$\neg r$

$\neg r$   $\left\{ \begin{array}{l} \text{conclusion} \end{array} \right.$

$(p \wedge q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r) \equiv$

$(p \rightarrow r) \wedge (q \rightarrow r) \equiv$

$(\neg p \vee r) \wedge (\neg q \vee r) \equiv$

$\neg(p \wedge q) \vee r \equiv$

| $\checkmark$<br>P | q | $\neg q$ | $\neg q$ | $\neg q$ | $\checkmark^a$<br>$P \rightarrow \neg q$ | $\checkmark^b$<br>$\neg q \rightarrow \neg q$ | $(P \wedge q \wedge b)$<br>$\neg q \rightarrow \neg q$ |
|-------------------|---|----------|----------|----------|------------------------------------------|-----------------------------------------------|--------------------------------------------------------|
| 1                 | 1 | 0        | 0        | 0        | 0                                        | 1                                             | 1                                                      |
| 1                 | 0 | 1        | 1        | 1        | 1                                        | 0                                             | 0                                                      |
| 0                 | 1 | 0        | 0        | 0        | 0                                        | 1                                             | 0                                                      |
| 0                 | 0 | 1        | 1        | 1        | 1                                        | 1                                             | 1                                                      |

Therefore, the argument is valid.

Q. PT w/o truth table

$$\begin{aligned}
 1) (P \rightarrow q) \wedge \neg q &\equiv \neg(P \vee q) \\
 (P \rightarrow q) \wedge \neg q &\equiv (\neg P \vee q) \wedge \neg q, \quad P \rightarrow q \equiv \neg P \vee q \\
 &\equiv (\neg q \vee \neg q) \vee \neg P \\
 &\equiv (\neg q \wedge \neg P) \vee (\neg q \wedge \neg q), \text{ distributive prop.} \\
 &\equiv (\neg q \wedge \neg P) \vee \neg q, \text{ negation law} \\
 &\equiv (\neg q \wedge \neg P), \text{ identity law} \\
 &\equiv \neg(P \vee q), \text{ commutative law \& de-Morgan's law.} \\
 &\equiv \text{RHS}
 \end{aligned}$$

Hence, proved

$$\begin{aligned}
 2) P \rightarrow (q \rightarrow r) &\equiv P \rightarrow (\neg q \vee r) \\
 &\equiv (P \wedge q) \rightarrow r
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} \equiv P \rightarrow (q \rightarrow r) &\equiv P \rightarrow (\neg q \vee r) \\
 &\equiv \neg P \vee (\neg q \vee r), \quad P \rightarrow q \equiv \neg P \vee q \\
 &\equiv (\neg P \vee \neg q) \vee r, \text{ commutative prop.} \\
 &\equiv \neg(P \wedge q) \vee r, \text{ de-morgan's law}
 \end{aligned}$$



$$\equiv (p \wedge q) \rightarrow r, \neg p \vee r \equiv p \rightarrow r$$

$$\equiv \text{RHS, hence proved}$$

$$3) (p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$\text{LHS} \equiv (p \rightarrow q) \vee (p \rightarrow r) \equiv (\neg p \vee q) \vee (\neg p \vee r)$$

$$\equiv (\neg p \vee \neg p) \vee (q \vee r), \text{ associative law.}$$

$$\equiv \neg p \vee (q \vee r), \text{ idempotent law.}$$

$$\equiv p \rightarrow (q \vee r), \neg p \vee q \equiv p \rightarrow q.$$

$$\equiv \text{RHS, hence proved}$$

$$4) \neg[(\neg(p \vee q) \wedge r) \vee \neg q] \equiv q \wedge r$$

$$\text{LHS} \equiv \neg[(\neg(p \vee q) \wedge r) \vee \neg q] \equiv \neg[(\neg p \wedge \neg q \wedge r) \vee \neg q], \text{ de-morgan's law.}$$

$$\equiv \neg[\neg(p \vee q) \wedge r] \wedge q, \text{ de-morgan.}$$

$$\equiv (p \vee q) \vee \neg r \wedge q$$

$$\equiv [(p \vee q) \wedge q] \vee (q \wedge \neg r)$$

$$\equiv [q \vee (q \wedge \neg r)] \wedge q$$

$$\equiv q \vee q \wedge (q \vee \neg r)$$

$$\equiv q \wedge (q \vee \neg r)$$

$$\equiv q$$



$$5) (p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \equiv \neg p \wedge q \quad (p \wedge q) \equiv$$

$$(p \vee q) \wedge [(\neg p \wedge \neg p) \wedge q] \equiv (p \vee q) \wedge (\neg p \wedge q)$$

$$\equiv ((p \vee q) \wedge \neg p) \wedge q$$

$$\equiv [(p \wedge \neg p) \vee (q \wedge \neg p)] \wedge q$$

$$\equiv [F \vee (q \wedge \neg p)] \wedge q$$

$$\equiv (q \wedge \neg p) \wedge q$$

$$\equiv \neg p \wedge q \equiv \text{RHS}$$

Hence, proved

$$\neg p \wedge q \equiv [p \vee (\neg p \wedge (\neg p \wedge q))] \quad (p \wedge q) \equiv$$

$$6) (\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \equiv r \quad [p \vee (\neg p \wedge (\neg p \wedge q))] \quad (p \wedge q) \equiv$$

$$\text{LHS} \equiv (\neg p \wedge (\neg q \wedge r)) \vee (q \wedge r) \vee (p \wedge r)$$

$$\equiv ((\neg p \wedge \neg q) \wedge r) \vee r \wedge (p \vee q), \text{ distributive \& commutative}$$

$$\equiv (\neg(p \vee q) \wedge r) \vee [r \wedge (p \vee q)], \text{ de-morgan's law}$$

$$\equiv r \wedge [\neg(p \vee q) \vee (p \vee q)], \text{ distributive law}$$

$$\equiv r \wedge T, \text{ negation law}$$

$$\equiv r, \text{ identity law}$$

$$\equiv \text{RHS, hence, proved}$$

| $\neg p$ | $p$ | $q$ | $r$ | $p \rightarrow q$ | $(p \rightarrow q) \wedge \neg q$ | $p \rightarrow \neg p$ |
|----------|-----|-----|-----|-------------------|-----------------------------------|------------------------|
| 0        | 1   | 1   | 0   | 1                 | 0                                 | 1                      |
| 0        | 1   | 0   | 1   | 0                 | 0                                 | 1                      |
| 1        | 0   | 1   | 0   | 1                 | 0                                 | 1                      |
| 1        | 0   | 0   | 1   | 1                 | 1                                 | 1                      |

## INFERENCE RULES

| INFERENCE RULE                                                     | LOGICAL IMPLICATION                                                          | NAME OF RULE           |
|--------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------|
| 1. $\frac{P \quad P \rightarrow q}{q}$                             | $[P \wedge (P \rightarrow q)] \rightarrow q$                                 | Modus ponens           |
| 2. $\frac{P \rightarrow q \quad \neg q}{\neg P}$                   | $[(P \rightarrow q) \wedge \neg q] \rightarrow \neg P$                       | Modus tollens          |
| 3. $\frac{P \rightarrow q \quad q \rightarrow r}{P \rightarrow r}$ | $[(P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (P \rightarrow r)$ | Hypothetical syllogism |
| 4. $\frac{P \quad q}{P \wedge q}$                                  | $[P \wedge q] \rightarrow (P \wedge q)$                                      | conjunction            |
| 5. $\frac{P \vee q \quad \neg P}{q}$                               | $[(P \vee q) \wedge \neg P] \rightarrow q$                                   | disjunctive syllogism  |
| 6. $\frac{P \wedge q}{P}$                                          | $P \wedge q \rightarrow P$                                                   | simplification         |
| 7. $\frac{P}{P \vee q}$                                            | $P \rightarrow (P \vee q)$                                                   | Addition               |
| 8. $\frac{P \vee q \quad \neg P \vee r}{q \vee r}$                 | $[(P \vee q) \wedge (\neg P \vee r)] \rightarrow (q \vee r)$                 | Resolution             |

Q. Rita is baking a cake — (refer back)

$$\begin{array}{l}
 P \\
 P \rightarrow \neg q \\
 \neg q \rightarrow \neg r \\
 \hline
 \neg r
 \end{array}$$

Symbolic Logic

SN.

STEPS

REASON

1

P

premise

2

$P \rightarrow \neg q$

premise

3

$\neg q \rightarrow \neg r$

premise

4

$P \rightarrow \neg r$

steps 2 & 3, hypothetical syllogism

5

$\neg r$

steps 1 & 4, modus ponens.

Q2 It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a trip and if we take a trip then we will be home by sunset. Therefore, we will be home by sunset. Establish the validity of the argument.

P: It is sunny this afternoon

q: It is colder than yesterday

r: we will go swimming

s: we will take a trip

$$\begin{array}{l}
 q \\
 \hline
 p \vee q
 \end{array}$$

$$\begin{array}{l}
 p \vee q \\
 r \vee q \\
 \hline
 r \vee p
 \end{array}$$

$t$ : we will be home by sunset.

Argument

$$\begin{array}{l} 7p \wedge q \\ 8p \rightarrow p \\ 78 \rightarrow s \\ s \rightarrow t \\ \hline t \end{array}$$

| SNO. | STEPS              | REASON                              |
|------|--------------------|-------------------------------------|
| 1.   | $7p \wedge q$      | premise                             |
| 2.   | $8 \rightarrow p$  | premise                             |
| 3.   | $78 \rightarrow s$ | premise                             |
| 4.   | $s \rightarrow t$  | premise                             |
| 5.   | $78 \rightarrow t$ | steps 3 & 4, hypothetical syllogism |
| 6.   | $7p$               | step 1, simplification              |
| 7.   | $78$               | steps 2 & 6, modus tollens          |
| 8.   | $t$                | steps 7 & 5, modus ponens           |

Therefore, the argument is valid.

Q3. If you send me an email message, then I will finish writing the program. If you don't send me an email message, then I will go to sleep early, and if I go to sleep early, then I will wake up feeling refreshed leading to the conclusion, if I don't finish writing the program, then I will wake up feeling refreshed. Establish the validity of the argument.

Ans:

$p$ : You send me an email message

$q$ : I will finish writing the program

$r$ : I will go to sleep early



$S$ : I will wake up feeling refreshed.

### Arguments

$$P \rightarrow Q$$

$$\neg P \rightarrow R$$

$$R \rightarrow S$$

---


$$\neg Q \rightarrow S$$

| SNO. | STEPS                       | REASON                              |
|------|-----------------------------|-------------------------------------|
| 1    | $P \rightarrow Q$           | premise                             |
| 2    | $\neg P \rightarrow R$      | premise                             |
| 3    | $R \rightarrow S$           | premise                             |
| 4    | $\neg P \rightarrow S$      | steps 2 & 3, hypothetical syllogism |
| 5    | $\neg Q \rightarrow \neg P$ | step 1, contrapositive              |
| 6    | $\neg Q \rightarrow S$      | steps 4 & 5, hypothetical syllogism |

$\therefore$  The argument is valid

Q4 If horses or cows eat grass, then mosquito is the national bird. If the mosquito is the national bird then peanut butter tastes good on hot dogs. But peanut butter tastes horrible on hot dogs, therefore the cow didn't eat grass. Establish the validity of the argument.

Ans:  $P$ : Horses ~~or cows~~ eat grass  $S$ : cows eat grass  
 $Q$ : Mosquito is the national bird.  
 $R$ : Peanut butter tastes good on hot dogs.

# Arguments:-

$$\begin{array}{l} SVP \rightarrow q \\ q \rightarrow r \\ \hline \neg r \\ \hline \neg S \end{array}$$

| SNO.                                                  | STEPS                  | REASON                              |
|-------------------------------------------------------|------------------------|-------------------------------------|
| 1                                                     | $SVP \rightarrow q$    | premise                             |
| 2                                                     | $q \rightarrow r$      | premise                             |
| 3.                                                    | $\neg r$               | premise                             |
| 4.                                                    | $SVP \rightarrow r$    | steps 1 & 2, hypothetical syllogism |
| 5.                                                    | $\neg (SVP)$           | steps 3 & 4, modus tollens          |
| 6                                                     | $\neg S \wedge \neg P$ | De-Morgan's law.                    |
| <u><math>\therefore</math> The argument is valid.</u> |                        |                                     |
| 7                                                     | $\neg S$               | step 6, simplification.             |

$\therefore$  argument is valid.

argument is valid

Q5 If Mohan is a lawyer then he is ambitious. If Mohan is an easy-riser then he does not like idlis. If Mohan is ambitious, then he is an easy-riser. Then if Mohan is a lawyer, then he did not like idlis. Show that the argument is valid.

$p$ : Mohan is a lawyer

$q$ : Mohan is ambitious

$r$ : Mohan is an easy-riser

$s$ : Mohan likes idlis.

Arguments.

$p \rightarrow q$

$r \rightarrow \neg s$

$q \rightarrow r$

Assume  
 $p$

$\neg s$

| SNO | STEPS                  | REASON                              |
|-----|------------------------|-------------------------------------|
| 1.  | $p \rightarrow q$      | premise                             |
| 2.  | $r \rightarrow \neg s$ | premise                             |
| 3.  | $q \rightarrow r$      | premise                             |
| 4.  | $p$                    | premise                             |
| 5.  | $p \rightarrow r$      | steps 1 & 3, hypothetical syllogism |
| 6.  | $p \rightarrow \neg s$ | steps 5 & 2, hypothetical syllogism |
| 7.  | $\neg s$               | steps 4 & 6, modus ponens           |

$\therefore$  argument is valid

Q.6 If Rachel gets the supervisor's position and works hard, then she will get a raise. If she gets the raise, then she will buy a new car. She has not purchased a new car, therefore either Rachel did not get the supervisor's position or she did not work hard. Show that the conclusion follows the premise.

p: Rachel gets the supervisor's position

q: Rachel works hard.

r: Rachel will get a raise

s: Rachel buys a new car

Arguments.

$p \wedge q \rightarrow r$

$r \rightarrow s$

$\neg s \rightarrow \neg (p \wedge q)$

$\neg p \vee \neg q$

| SNO. | STEPS                      | REASON                              |
|------|----------------------------|-------------------------------------|
| 1.   | $p \wedge q \rightarrow r$ | premise                             |
| 2.   | $r \rightarrow s$          | premise                             |
| 3.   | $\neg s$                   | premise                             |
| 4.   | $p \wedge q \rightarrow s$ | steps 1 & 2, hypothetical syllogism |
| 5.   | $\neg (p \wedge q)$        | steps 3 & 4, modus tollens.         |
| 6.   | $\neg p \vee \neg q$       | de-morgan's law                     |

$\therefore$  argument is valid

follows the premise  $\therefore$



Q7 IF it does not rain or if it is not foggy, then the sailing race will be held and the life saving demonstration will go on. If the sailing race is held the trophy will be awarded. And the trophy was not awarded implied the conclusion "it rained". Use the rules of inference to test the validity of the argument.

p: It rains

q: It's foggy

r: sailing race is held

s: life saving demonstration will go on.

t: trophy will be awarded

### Arguments

$$\neg p \vee \neg q \rightarrow r \wedge s$$

$$r \rightarrow q$$

$$\neg t$$

$$\underline{p}$$

| SNO. | STEPS                                                                    | REASON                     |
|------|--------------------------------------------------------------------------|----------------------------|
| 1    | $\neg p \vee \neg q \rightarrow r \wedge s$                              | premise                    |
| 2    | $r \rightarrow q$                                                        | premise                    |
| 3    | $\neg t$                                                                 | premise                    |
| 4    | $\neg r$                                                                 | step 2 & 3, modus tollens. |
| 5    | $\neg r \vee \neg s$                                                     | step 4, addition           |
| 6    | $\neg(r \wedge s)$                                                       | de-morgan's law            |
| 7    | $\neg(\neg p \vee \neg q)$<br><del><math>\neg p \vee \neg q</math></del> | step 1 & 6, modus tollens  |
| 8    | $p \wedge q$                                                             | de-morgan's law            |
| 9    | $p$                                                                      | simplification.            |

$\therefore$  argument is valid

Q8. If today is Tuesday, I have a test of Mathematics or Economics. If my ~~Physics~~ <sup>Economics</sup> prof. is sick, I did not have a test of Economics. Today is Tuesday and my Economics prof. is sick, therefore I have a test of Mathematics. ST the argument is valid.

Arguments.

P: Today is Tuesday

q: I have a test of Mathematics

r: I have a test of Economics

s: My Economics prof. is sick

$$P \rightarrow q \vee r$$

$$s \rightarrow \neg r$$

$$P \wedge s$$

$$q$$

SNO.

STEPS

REASON

1.

$$P \rightarrow q \vee r$$

premise

2.

$$s \rightarrow \neg r$$

premise

3.

$$P \wedge s$$

premise

4.

$$s$$

simplification, step 3.

5.

$$q \vee r$$

steps 1 & 4, modus ponens

6.

$$\neg r$$

steps 2 & 4, modus ponens

7.

$$q$$

steps 5 & 6, disjunctive syllogism

$\therefore$  argument is valid

Q.9. If Dominic goes to the race track, then Helen will be mad. If Ralph plays cards all night, then Kasmala will be mad. If either Helen or Kasmala get mad then Veronica will be notified.

Veronica has not heard from either of these clients, i.e., Veronica is not notified, consequently Dominic did not make it to the race track and Ralph didn't play cards all night. Establish the validity of the statement.

### Arguments

p: Dominic goes to the race track

$p \rightarrow q$

q: Helen will be mad

$r \rightarrow s$

r: Ralph plays cards all night

$q \vee s \rightarrow t$

s: Kasmala will be mad

$\neg t$

t: Veronica will be notified

$\neg p \wedge \neg r$

| SNO. | STEPS                    | REASON                     |
|------|--------------------------|----------------------------|
| 1.   | $p \rightarrow q$        | premise                    |
| 2.   | $r \rightarrow s$        | premise                    |
| 3.   | $q \vee s \rightarrow t$ | premise                    |
| 4.   | $\neg t$                 | premise                    |
| 5.   | $\neg (q \vee s)$        | steps 3 & 4, modus tollens |
| 6.   | $\neg q \wedge \neg s$   | de Morgan's law            |
| 7.   | $\neg q$                 | step 6, simplification     |
| 8.   | $\neg p$                 | steps 1 & 7, modus tollens |
| 9.   | $\neg s$                 | step 6, simplification     |
| 10.  | $\neg r$                 | step 2 & 9, modus tollens  |
| 11.  | $\neg p \wedge \neg r$   | steps 8 & 10, conjunction  |

$\therefore$  argument is valid

5. write the converse, inverse and contrapositive

If  $-1 < 3$  and  $3+7=10$ , then  $\sin\left(\frac{3\pi}{2}\right) = -1$

$p: -1 < 3$

$q: 3+7=10$

$r: \sin\left(\frac{3\pi}{2}\right) = -1$

Given statement is  $(p \wedge q) \rightarrow r$

converse:  ~~$\neg(p \wedge q)$~~   $r \rightarrow (p \wedge q)$

If  $\sin\left(\frac{3\pi}{2}\right) = -1$ , then  $-1 < 3$  and  $3+7=10$ .



Inverse:  $\neg(p \wedge q) \rightarrow \neg x$

$\neg p \vee \neg q \rightarrow \neg x$  (de-morgan's law)

If  $-1 \geq 3$  or  $3+7 \neq 10$  then  $\sin(\frac{3\pi}{2}) \neq -1$

Contrapositive:  $\neg x \rightarrow \neg(p \wedge q) \equiv \neg x \rightarrow \neg p \vee \neg q$

If  $\sin(\frac{3\pi}{2}) \neq -1$ , then  $-1 \geq 3$  or  $3+7 \neq 10$ .

(ii) If I finish my homework, before dinner and it does not rain, then I will go to the game.

p: I finish my homework before dinner

q: It rains

x: I will go to the game.

Given statement,  $(p \wedge \neg q) \rightarrow x$

Converse:  $x \rightarrow (p \wedge \neg q)$

If I ~~will~~ go to the game, then I will finish my homework before dinner and it does not rain.

Inverse:  $\neg(p \wedge \neg q) \rightarrow \neg x$

$\neg p \vee q \rightarrow \neg x$  (de-morgan's law and double negation)

If I do (not) finish my homework before dinner or it rains, then I will not go to the game!

Contrapositive:  $\neg x \rightarrow \neg(p \wedge \neg q)$

$\neg x \rightarrow \neg p \vee q$

If I don't go to the game, then I didn't finish my homework before dinner or it rains.

(iii) I go to the beach, whenever it is a sunny summer day.

$p$ : I go to the beach

$q$ : It is a sunny summer day.

Given statement is  $p \rightarrow q$

If it is a sunny summer day, then I will go to the beach

$p \rightarrow q$   
if  $p$ , then  $q$   
 $p$  implies  $q$   
 $p$  only if  $q$

converse:  $q \rightarrow p$

If I go to the beach, then it is a sunny summer day

$q$  whenever  $p$ .  
 $q$  follows from  $p$ .  
 $q$  when  $p$ .

inverse:  $\neg p \rightarrow \neg q$

If it is not a sunny summer day then I will not go to the beach.

$q$  is necessary for  $p$   
 $q$  provided that  $p$ .

contrapositive:  $\neg q \rightarrow \neg p$

If I do not go to the beach then it is not a sunny summer day.

(iv) The hometeam wins whenever it is raining.

$p$ : It is raining

$q$ : The hometeam wins.

Given,  $p \rightarrow q$

converse

~~inverse~~:  $q \rightarrow p$

If the hometeam wins, then it is raining.

inverse

~~converse~~:  $\neg p \rightarrow \neg q$

If it is not raining then the hometeam did not win.

contrapositive:  $\neg q \rightarrow \neg p$

If the hometeam does not win, then it is not raining.

(v) A positive integer is a prime, only if it has no divison other than 1 and itself.

$P$ : a positive integer is a prime.

$q$ : <sup>a +ve integer</sup> has ~~no~~ divison other than 1 and itself.

Given,  $P \rightarrow q$

Converse:  $q \rightarrow P$

If a positive integer has no divison other than 1 and itself then it is a ~~per~~ prime.

Inverse:  $\neg P \rightarrow \neg q$

If a positive integer is not a prime, then it has divison other than 1 and itself.

Contrapositive:  $\neg q \rightarrow \neg P$

If a positive integer has divison other than 1 and itself then it is not a prime.

=====

(vi) If it snows today, I will ski tomorrow.

$P$ : It snows today

$q$ : I will ski tomorrow.

Given,  $P \rightarrow q$

~~Converse~~  $q \rightarrow P$

If I ~~will~~ ski tomorrow, then it will snow today.

~~Inverse~~  $\neg q \rightarrow \neg P$

~~Converse~~



Inverse:  $\neg p \rightarrow \neg q$

If it does not snow today, I will not ski tomorrow

Contrapositive:  $\neg q \rightarrow \neg p$ .

If I do not ski tomorrow, it did not snow today =====.

Q. "Randy works hard". If Randy works hard, then "he is a dull boy" and if Randy is a dull boy, then "he will not get the job". Imply the conclusion, "Randy will not get the job". Check the validity.

p: Randy works hard

q: He is a dull boy

r: He will get the job

Arguments

$$\begin{array}{l} p \\ p \rightarrow q \\ q \rightarrow \neg r \\ \hline \neg r \end{array}$$

S. No.

Steps

Reason.

1.

$$p \rightarrow q$$

premise

2.

$$q \rightarrow \neg r$$

premise

3.

$$p \rightarrow \neg r$$

1 & 2, hypothetical syllogism

4.

$$p$$

premise.

5.

$$\neg r$$

3 & 4, modus ponens

Hence, valid



25/04/25

## MODULE - 2

### METHOD OF PROVING THEOREMS

1. DIRECT METHOD
2. INDIRECT METHOD (Proof by contraposition)
3. PROOF BY CONTRADICTION
4. PROOF BY COUNTEREXAMPLE

Q 1. Give a direct proof of the theorem "if 'n' is an odd integer, then 'n<sup>2</sup>' is odd".

Assume 'n' is odd.

$\therefore n = 2k + 1$ , where k is some integer.

$$\text{Then } n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$

$= 2(2k^2 + 2k) + 1$ , which is an odd integer.

Hence, the theorem.



2. Give a direct proof "if m and n are both perfect squares, then mn is also a perfect square".

Ans: let  $m = s^2$  and  $n = t^2$

$$\therefore mn = s^2 t^2 = (s \cdot t)^2$$

$$= (st)(st), \text{ associativity}$$

$$= (st)^2$$

Hence, the theorem.

3. "Sum of two odd integers is even"

let  $k = 2a + 1$  and  $m = 2b + 1$

$$\begin{aligned}\text{then, } k + m &= 2a + 1 + 2b + 1 \\ &= 2(a) + 2b + 2 \\ &= 2(a + b + 1), \text{ which is even.}\end{aligned}$$

Hence, the theorem

4. "Sum of two even integers is even"

let  $k = 2a$  and  $m = 2b$ .

$$\begin{aligned}\text{then } k + m &= 2a + 2b \\ &= 2(a + b), \text{ which is even.}\end{aligned}$$

Hence, the theorem

5. "Square of an even no. is an even no."

let  $k = 2a$

$$\text{then, } k^2 = (2a)^2$$

$$= 2a \cdot 2a$$

$$= 2 \cdot 2 \cdot a \cdot a$$

$$= 4a^2$$

$$= 2(2a^2), \text{ which is even.}$$

Hence, the theorem

6. "Product of two odd no.s is odd"

let  $k = 2a + 1$  and  $m = 2b + 1$ .

$$\text{Then } k \cdot m = (2a + 1)(2b + 1)$$

$$= 4ab + 2a + 2b + 1$$

$$= 2(2ab + a + b) + 1, \text{ which is odd}$$

Hence, the theorem

7. PT "if  $m+n$  and  $n+p$  are even integers,  $m, n, p$  are integers, then  $m+p$  is even."

Ans: let  $m+n = 2k$  and  $n+p = 2t$

then,  $m+n + n+p = 2k + 2t$

$$m+2n+p = 2(k+t), \text{ ~~which~~$$

$$m+p = 2[k+t-n], \text{ which is even.}$$

Hence, the theorem

Proof by contradiction

Q.1) ~~Root~~  $\sqrt{2}$  is irrational

If possible, we assume that  $\sqrt{2}$  is rational:

i.e.,  $\sqrt{2} = \frac{p}{q}$ ,  $q \neq 0$ ,  $p$  and  $q$  have no common factor

$$\sqrt{2} = \frac{p}{q}$$

squaring,  $2 = \frac{p^2}{q^2}$

$$2q^2 = p^2$$

$$p^2 = 2q^2, \text{ which is even}$$

$$\text{i.e., } p^2 \text{ is even}$$

$$\therefore p \text{ is even}$$

$$\text{i.e., } \boxed{p = 2k}, \text{ where } k \text{ is an integer.}$$

$$\text{substitute in } p^2 = 2q^2$$

$$(2k)^2 = 2q^2$$

$$2q^2 = 4k^2$$

$$q^2 = 2k^2, \text{ which is even}$$

$$\text{i.e., } q^2 \text{ is even.}$$

$$\therefore q \text{ is also even.}$$



$$\therefore \underline{q = 2m}$$

Here,  $q$  and  $p$  are even.

$\therefore 2$  divides both  $p$  and  $q$ , which is a contradiction that  $p/q$  is in lowest form.

$\therefore \underline{\sqrt{2}}$  is irrational.

Q2. Give a proof by contradiction of the theorem, "if  $3n+2$  is odd, then  $n$  is odd."

Assume that  $n$  is even, i.e.,  $n = 2k$

$$\text{Then, } 3n+2 = 3(2k)+2$$

$= 2[3k+1]$ , which is even, which is a contradiction.

$\therefore$  if  $3n+2$  is odd, then  $n$  is odd.

Q3. Give a proof by contradiction "if  $3n+2$  is even then  $n$  is even."

Assume that  $n$  is odd, i.e.,  $n = 2k+1$ .

$$\text{then, } 3n+2 = 3(2k+1)+2$$

$$= 3(2k)+3+2$$

$$= 2(3k+1)+2+1$$

$= 2(3k+2)+1$ , which is odd, which is a contradiction.

$\therefore$  if  $3n+2$  is even, then  $n$  is even.

Q4. PT "if  $a^2$  is divisible by 3, then  $a$  is divisible by 3".

Assume that  $a$  is not divisible by 3.

$$\text{then, } a = 3k+1$$

$$\therefore a^2 = (3k+1)^2 = 9k^2 + 6k + 1$$

$= 3(3k^2 + 2k) + 1$ , which is not divisible by 3, contradiction.

$\therefore$  if  $a^2$  is divisible by 3, then  $a$  is divisible by 3.



Q5. Give a proof by contradiction,

"For all integer 'n', if  $n^2$  is odd, then n is odd."

Assume that n is even,

$$n = 2k$$

$$\text{then, } n^2 = (2k)^2$$

$$= 4k^2 = 2(2k^2), \text{ which is even, contradiction.}$$

Hence, the theorem

~~Q6.~~ Proof by contraposition (indirect method)

Q1. If 'n' is an integer, and  $5n+2$  is odd, then n is odd.

Assume 'n' is even,

$$\therefore n = 2k$$

$$\text{then, } 5n+2 = 5(2k)+2$$

$$= 2(5k+1), \text{ which is even}$$

i.e., n is even, then  $5n+2$  is even.

i.e.,  $5n+2$  is odd, then n is odd.

Q2. 'n' is an integer, and  $n^3+5$  is odd, then n is even.

Assume that n is odd,

$$\therefore n = 2k+1$$

$$\text{then, } n^3+5 = (2k+1)^3+5$$

$$= 8k^3+1+3 \times 2k(2k+1)+5$$

$$= 8k^3+1+12k^2+6k+5$$

$$= 8k^3+6+12k^2+6k$$

$$= 2(4k^3+3k^2+3k+3),$$

which is even.

i.e., n is odd, then  $n^3+5$  is even.

then  $n^2 + 5$  is odd, then  $n$  is even

Q3. PT  $n$  is a positive integer, then  $n$  is odd iff  $5n+6$  is odd.

First, we assume that  $n$  is odd, then we prove that  $5n+6$  is odd.

$$\begin{aligned}\text{let } n &= 2k+1, \text{ then } 5n+6 = 5(2k+1)+6 \\ &= 10k+5+6 \\ &= 10k+10+1 \\ &= 2(5k+5)+1, \text{ which is odd}\end{aligned}$$

Conversely, we assume that  $n$  is even,

$$\text{i.e., } n = 2k, \text{ then } 5n+6 = 2(5k+3), \text{ which is even.}$$

i.e.,  $n$  is even, then  $5n+6$  is even.

$\Rightarrow 5n+6$  is odd, then  $n$  is odd.

Hence, the theorem

Q4. PT "if  $n$  is a +ve integer, then  $n$  is even iff  $7n+4$  is even."

First, we assume that  $n$  is even, then we PT  $7n+4$  is even.

$$\Rightarrow n = 2k$$

$$\text{then, } 7n+4 = 2(7k+2), \text{ which is even.}$$

Conversely, we assume that  $n$  is odd,

$$\Rightarrow n = 2k+1$$

$$\begin{aligned}\text{then, } 7n+4 &= 7(2k+1)+4 \\ &= 14k+11 \\ &= 2(7k+5)+1, \text{ which is odd.}\end{aligned}$$

i.e.,  $n$  is odd, then  $7n+4$  is odd

$\Rightarrow 7n+4$  is even, then  $n$  is even

Hence, the theorem



Q5. ST 'if  $n$  is an integer, and  $n^3+5$  is odd, then  $n$  is even', using ① by contradiction  
② by contraposition

① We assume that  $n$  is odd,

$$\text{then, } n = 2k+1,$$

$$\Rightarrow n^3+5 = (2k+1)^3+5$$

$$= (8k^3+1+3k \times 2(2k+1))+5$$

$$= 2(4k^3+3+3k(2k+1)), \text{ which is even}$$

(Since,  $n^3$  is odd, 5 is odd, odd+odd is even)

Hence, the theorem

② We assume that  $n$  is odd,

$$\text{then } n = 2k+1$$

$$\Rightarrow n^3+5 \text{ is even}$$

$\therefore$  If  $n$  is odd, then  $n^3+5$  is even

$\Rightarrow n^3+5$  is odd then  $n$  is even.

### PIGEON-HOLE PRINCIPLE

If ' $m$ ' pigeons occupy ' $n$ ' pigeonholes,  $m > n$ , then atleast one pigeonhole will contain more than one pigeon.

If  $k$  is a +ve integer and  $(k+1)$  or more objects are placed into  $k$  boxes, then there is atleast 1 box containing two or more of the objects.

If ' $m$ ' pigeons are put in ' $n$ ' pigeonholes,  $m > n$ , then one of the pigeonhole will contain

atleast  $\left\lfloor \frac{m-1}{n} \right\rfloor + 1$

Q1. ST in a group of 13 children, there must be atleast two children who were born in same month.

Then  $m = 13 = \text{pigeon}$   
 $n = 12 = \text{pigeonhole}$

$$\therefore \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{13-1}{12} \right\rfloor + 1 = \underline{\underline{2}}$$

$\therefore$  2 children were born in same month

Q2. Given a group of 100 people, at minimum how many people were born in same month?

Then,  $m = \text{pigeon} = 100$   
 $n = \text{pigeonhole} = 12$

$$\therefore \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{100-1}{12} \right\rfloor + 1 = 8 + 1 = \underline{\underline{9}}$$

$\therefore$  minimum 9 people were born in same month

Q3. ST in any group of 8 people, atleast 2 have birthdays which fall on the same days.

$m = 8, n = 7$

$$\therefore \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{8-1}{7} \right\rfloor + 1 = \underline{\underline{2}}$$

$\therefore$  2 have same birthday.

Q4. ST in any 367 people, atleast 2 have same bday.

$m = 367, n = 365$

$$\therefore \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{367-1}{365} \right\rfloor + 1 = \underline{\underline{2}}$$

$\therefore$  Atleast 2 have same bdays.



Q5. In a group of 27 English words, there must be at least 2 begin with the same letter.

$$m = 27, n = 26$$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{27-1}{26} \right\rfloor + 1 = \underline{\underline{2}}$$

$\therefore$  there are at least 2 beginning with same letter

26/04/25

Q6. How many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 point?

Ans:  $m = ?$ ,  $n = 101$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2$$

$$\left\lfloor \frac{m-1}{n} \right\rfloor = 1$$

$$\left\lfloor \frac{m-1}{101} \right\rfloor = 1 \Rightarrow m = \underline{\underline{102}}$$

$\therefore$  102 students must be in the class

Q7. What is the minimum no. of students required in an English class to be sure that at least 6 will receive the same grade, if there are 5 possible grades.

Ans:  $m = ?$ ,  $n = 5$

$$\left\lfloor \frac{m-1}{5} \right\rfloor + 1 = 6$$

$$\left\lfloor \frac{m-1}{5} \right\rfloor = 5$$

$$\underline{\underline{m = 26}}$$

Q8. what is the minimum no. of students each of whom comes from one of the 50 states who must be enrolled in the university to guarantee that they<sup>re</sup> are atleast 100 who come from the same state?

Ans:  $m = ?$   $n = 50$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 100$$

$$\left\lfloor \frac{m-1}{50} \right\rfloor = 99$$

$$m = \underline{\underline{4951}}$$

$$\begin{array}{r} 99 \\ \times 50 \\ \hline 4950 \end{array}$$

Q9. ST in any set of 6 classes, each meeting regularly once a week on a particular day of the week there must be 2 that meet on the same day, assuming that no classes are held on weekends.

Ans:  $m = 2$   $6$

$n = 5$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{6-1}{5} \right\rfloor + 1 = 1 + 1 = \underline{\underline{2}}$$

$\therefore$  minimum 2 <sup>classes</sup> meet on the same day.

Hence, proved

Q10. ST if there 30 students in a class then atleast 2 have last names that begin with the same letter.

$m = 30$  ,  $n = 26$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{30-1}{26} \right\rfloor + 1 = \underline{\underline{2}}$$

$\therefore$  atleast 2 students have last names with same beginning letter.

Hence, proved



Q11. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random w/o looking at them.

a) How many balls must she select to be ~~sure~~ sure of having atleast 3 balls of the same colour?

b) How many balls must she select to be sure of having atleast 3 blue balls?

Ans: a)  $m = ?$

$$n = 2$$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 3$$

$$\left\lfloor \frac{m-1}{2} \right\rfloor = 2$$

$$\Rightarrow \underline{\underline{m = 5}}$$

b) Minimum 13 balls must be selected to be sure of having atleast 3 blue balls. (worst-case 10 red + 3 blue)

Q12. A drawer contains a dozen brown socks and dozen black socks all unmatched. A man takes socks out at random in the dark.

a) How many socks must he take out to be sure that he has atleast 2 socks of the same colour?

b) How many socks must he take out to be sure that he has atleast 2 black socks?

Ans:

a)

$$m = ? , n = 2$$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2$$

$$\left\lfloor \frac{m-1}{2} \right\rfloor = 1 \Rightarrow \underline{\underline{m = 3}}$$

b) He must take out 14 socks minimum to get atleast 2 black socks.

Q13. An auditorium has a seating capacity of 1800. How many seats must be occupied to guarantee that atleast 2 people seated in the auditorium have the same first and last initials?

Ans:  $m = ?$ ,  $n = 676$ .

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2.$$

$$\left\lfloor \frac{m-1}{676} \right\rfloor + 1 = 2$$

$$\left\lfloor \frac{m-1}{676} \right\rfloor = 1 \Rightarrow \underline{m = 677}$$

Q14. 16 students in a college belong to 1 of 4 groups depending on the years in which they are expected to graduate. Each student must choose 1 of 21 different majors. How many students are ~~there~~ <sup>who</sup> there to assure that there are 2 students ~~that~~ <sup>who</sup> choose same major, in the same year.

$$m = ?$$
,  $n = 21 \times 4 = \underline{84}$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2$$

$$\left\lfloor \frac{m-1}{84} \right\rfloor = 1 \Rightarrow \underline{m = 85}$$

$\therefore$  minimum 85 students need to be there to assure that there are 2 students who have same major and year of graduation.

$$\begin{array}{r} 28 \\ 26 \\ \hline 526 \\ 676 \end{array}$$



Q15. There are 6 professors teaching the introductory dm class at university. The same final exam is given by all 6 professors. If the lowest possible score on the final is 0 and the highest possible score is 100. How many students must be there to guarantee that there are 2 students with the same prof. who earned the same final exam score?

Ans:

$$m = ? , n = 101 \times 6 = 606$$

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2$$

$$\left\lfloor \frac{m-1}{606} \right\rfloor = 1 \Rightarrow \underline{m = 607}$$

OR  $\frac{m-1}{n} = 1 \Rightarrow$

[There are 101 possible scores, since 0 and 100 are both included and 6 professors. So, there are  $6 \times 101 = 606$  possible professors-score pairs.]

$\therefore 606 + 1 = 607$  students are sufficient to guarantee that there are minimum 2 who ~~can~~ have same score and prof.

Q16. ST if any 4 no.s from 1 to 6 are chosen, then atleast 2 of them will add to 7.

$$[(1,6), (2,5), (3,4)]$$

$$m = 4 , n =$$

$\{1, 2, 3, 4, 5, 6\}$ , the subsets are  $\{(1,6), (2,5), (3,4)\}$

$$\underline{n=3}$$

$$\text{Then, } \left\lfloor \frac{m-1}{n} \right\rfloor + 1 = \left\lfloor \frac{4-1}{3} \right\rfloor + 1 = \underline{2}$$

Hence, proved

Q17. How many no.s must be selected from the set  $\{1, 3, 5, 7, 9, 11, 13, 15\}$  to guarantee that atleast 1 pair of these no.s add upto 16?

Ans:  $m = ?$ ,  $n = 8$   
 $\{(1, 15), (3, 13), (5, 11), (7, 9)\}$  is the possible subset.  
 $n = 4$ .

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 2^m$$

$$\left\lfloor \frac{m-1}{4} \right\rfloor = 0 \Rightarrow \underline{m = 5}$$

DERANGEMENT  
 It is an arrangement of objects in which none of the objects in its original position.

Eg:- 1, 2, 3

Derangements are 2, 3, 1 and 3, 1, 2  
 Total 2 derangements.

→ The no. of derangement of  $n$  elements is

$$D_n = n! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{(-1)^n}{n!} \right]$$

Q1 Find  $D_3$ .

$$D_3 = 3! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right] = 6 \left[ 1 - 1 + \frac{1}{2} - \frac{1}{6} \right]$$

$$= 6 \left[ \frac{3-1}{2} \right] = \underline{2}$$



Q2. Find the no. of ~~element~~ derangement of  $\{1, 2, 3, 4\}$ .

$$D_4 = 4! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

$$= 24 \left[ \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right] = 24 \left[ \frac{12 - 4 + 1}{24} \right]$$

$$= \underline{\underline{9}}$$

Q3. There are 8 ~~des~~ guest at a party. Each guest brings and each receives another gift in return. None are allowed to receive a gift they bought. How many ways are there to distribute gift?

$$D_8 = 8! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} \right]$$

$$= \underline{\underline{14833}}$$

Q4. What is the probability that none of 10 people receives the correct hat, if a hat-check person hands their hats back randomly?

$$P(\text{none gets hat back correctly}) = \frac{D_{10}}{10!}$$

$$= \frac{10! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} - \frac{1}{9!} + \frac{1}{10!} \right]}{10!}$$

$$= \underline{\underline{0.36788}}$$

Q5. A machine that inserts letters into envelopes goes haywire and inserts letters randomly into envelopes. What is the probability that, in a group of 100 letters:

a) No letter is put into the correct envelope

- b) Exactly 1 letter is put into the crct envelope.
- c) Exactly 98 letters are put into the crct envelopes.
- d) 99 letters are put into the correct envelopes.
- e) All letters are put into the correct envelopes.

Ans:

a)  $P(\text{no letter is put in correct envelope}) = 1 - \frac{D_{100}}{100!}$

[Total no. of cases =  $100!$ ]

$$= \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{1}{100!} \right]$$

b)  $P(\text{exactly 1 letter is put correctly}) = \frac{D_{99}}{100!} \times 100C_1$

$$= \frac{D_{99}}{99!}$$

$$= \frac{1}{100}$$

c)  $P(\text{exactly 98 letters are put correctly}) = \frac{D_2 \times 100C_{98}}{100!}$

$$= \frac{100!}{98! 2!} \times \frac{D_2}{100!}$$

$$= \frac{D_2 \times 100C_2}{100!}$$

$$= \frac{1}{100}$$

d)  $P(\text{exactly 99 letters are put correctly}) = \frac{D_1}{100!} = 0$

e)  $P(\text{all letters put correctly}) = \frac{1}{100!}$



Q8. How many onto functions are there from a set with 6 elements to a set with 3 elements?

→ let  $m$  and  $n$  be +ve integers with  $m \geq n$ , then the no. of onto functions from a set with  $m$  elements to a set with  $n$  elements is

$$n^m - nC_1(n-1)^m + nC_2(n-2)^m - \dots + (-1)^{n-1}nC_{n-1}1^m$$

Here,  $m=6$   
 $n=3$

$$\Rightarrow 3^6 - 3C_1(2)^6 + 3C_2(1)^6 - 3C_3(0)^6 = \left[ \frac{001}{211} \right] = (200)u$$

$$\Rightarrow 3^6 - \frac{3!}{2!1!}(2)^6 + \frac{3!}{2!1!}(1)^6 = \underline{\underline{540}} = \left[ \frac{001}{211} \right] = (200)u$$

Q9. How many ways are there to assign 5 different jobs to 4 different employees, if every employee is assigned atleast one job?

$m=5, n=4$

$$4^5 - 4C_1(3)^5 + 4C_2(2)^5 - 4C_3(1)^5$$

$$= 4^5 - 4(3)^5 + \frac{4!}{2!2!}(2)^5 - 4 \left[ \frac{001}{122} \right] = (8000)u$$

$$= 4^5 - 4(3)^5 + 6(2)^5 - 4 = \underline{\underline{240}} = \left[ \frac{001}{122} \right] = (10000)u$$

Q10. How many ways are there to can 8 distinct balls be distributed into 3 distinct box if each box must contain atleast one ball?

$m=8, n=3$

$$3^8 - 3C_1(2)^8 + 3C_2(1)^8 - 3C_3(0)$$

$$= 3^8 - 3(2)^8 + 3(1)^8 = \underline{\underline{5796}} u$$

$(10000)u + (10000)u + (20000)u =$

$(100000)u + (100000)u =$